

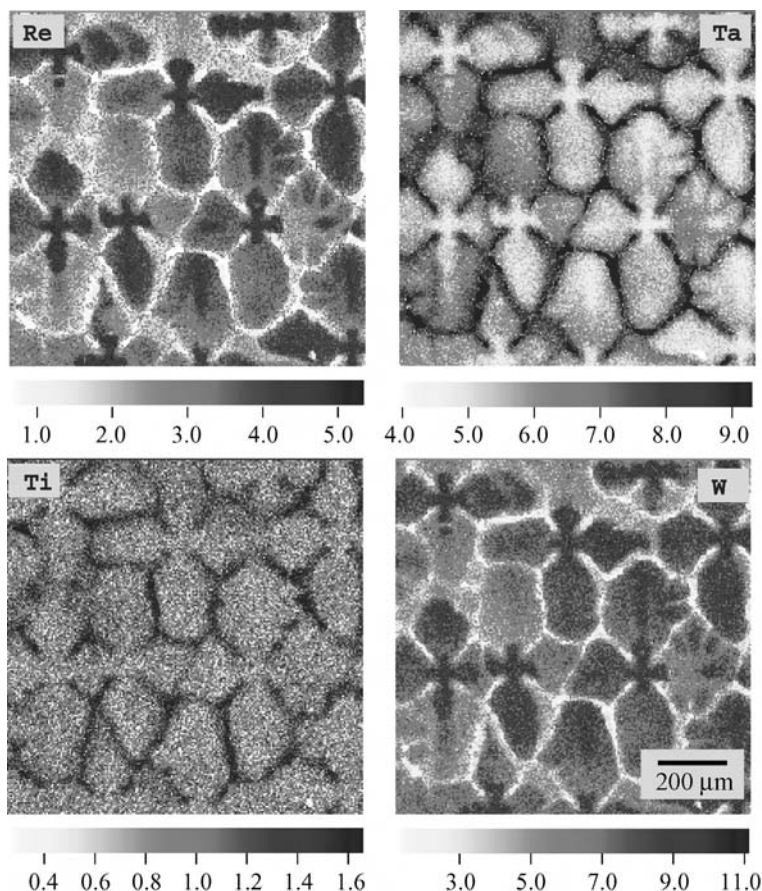


Fig. 3.23. Electron probe microanalysis (EPMA) maps of the transverse section of the as-cast structure of the CMSX-4 superalloy, which confirms the partitioning of W and Re to the dendrite core, and Ti and Ta to the interdendritic regions. (Courtesy of Mudith Karunaratne.)

for the experimental alloys reported in ref. [18] for which freckling data are available; see Figure 3.25. When the density difference, $\Delta\rho$, is evaluated at a supercooling of 20 K below the liquidus, a correlation is found with the average number of freckle chains found per blade sample. This provides strong evidence that the density inversion is responsible for the driving force for the freckling phenomenon.

3.1.4 *The influence of processing conditions on the scale of the dendritic structure*

During processing by directional solidification, the temperature gradient, G , and the withdrawal velocity, v , influence the scale of the dendritic structure which is formed. For example,

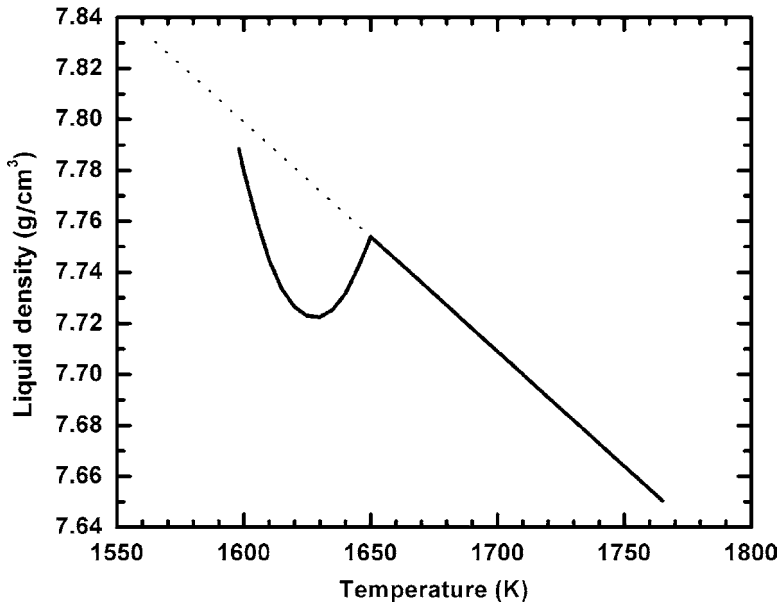
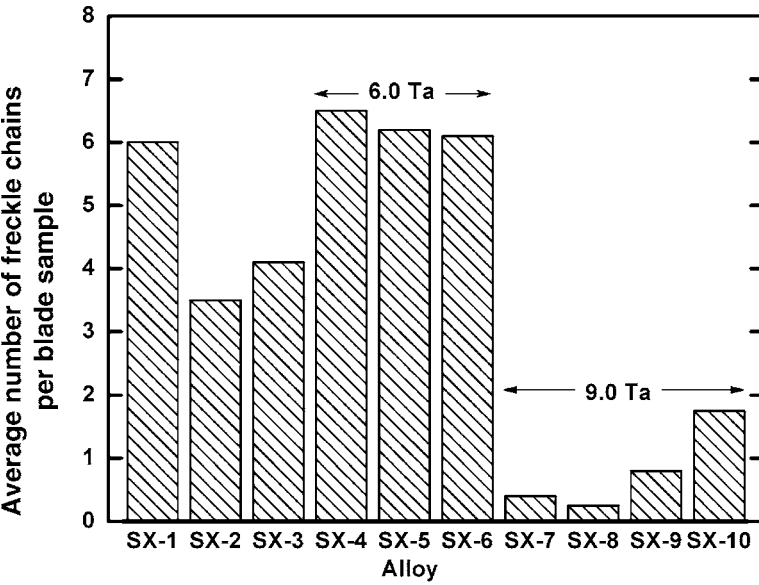


Fig. 3.24. Variation of the density of liquid CMSX-4 with temperature. The data beyond 1650 K are those recommended by ref. [21] – the dotted line represents an extrapolation below the liquidus. The solid line below 1650 K represents the calculated density of the liquid in the mushy zone, assuming equilibrium is attained; thus, the effect of solute partitioning is accounted for.

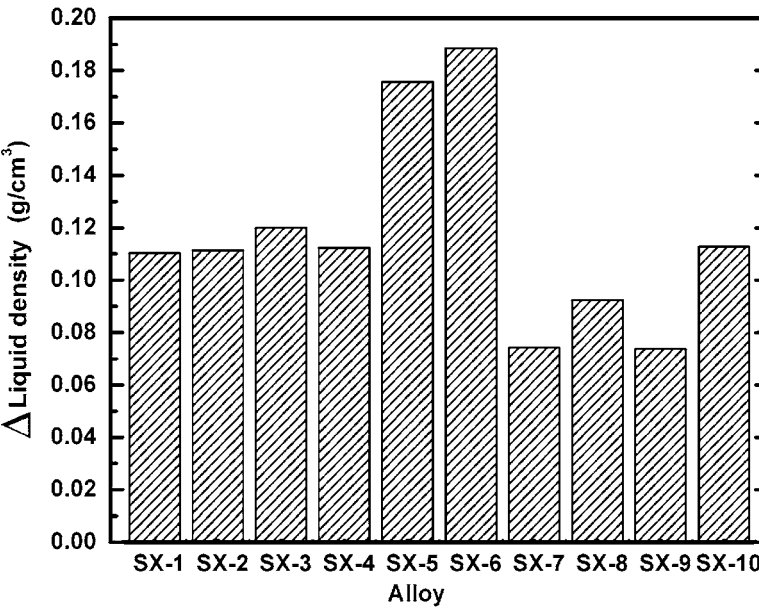
the use of liquid metal cooling is found to decrease the dendrite arm spacing at otherwise identical withdrawal velocity; see Figure 3.8. It is helpful to estimate the relative potencies of G and v on the scale of the microstructure to be expected, for example the primary dendrite arm spacing, L_p .

In order to do this, consider first the temperature interval, ΔT_0 , required for solidification to be completed. If the phase diagram is taken to be a binary one with a liquidus of slope m and a solid/liquid partitioning coefficient of k , then it can be demonstrated that ΔT_0 is given by $mc_0(k-1)/k$, where c_0 is the bulk composition of the alloy under consideration. The scale of the dendritic structure is then as depicted in Figure 3.26. The dendrites are assumed to possess a radius of curvature R at their tips, which are spaced a distance L_p apart. The mean height of the dendrite trunks is then $\Delta T_0/G$, where G is the temperature gradient; the thickness of the trunks at their roots is given the symbol λ , which is approximately one half of L_p . A steady-state is assumed – thus, the dendritic structure translates at a velocity v , and its shape is invariant as it does so. A first estimate of the radius of curvature of the tip of each dendrite is given by fitting a parabola to its shape [22]; from elementary geometry, it is then equal to the ratio of the square of the minor axis of the parabola (which is equal to λ) to the major axis ($\Delta T_0/G$) – so that $R = \lambda^2 G / \Delta T_0$.

For the solidification conditions pertinent to the superalloys, there are two contributions to the total undercooling ΔT . The first is due to the solutal undercooling, $\Delta T_{\text{solutal}}$, arising from the partitioning of solute; the second, $\Delta T_{\text{curvature}}$, is a consequence of the Gibbs–Thomson



(a)



(b)

Fig. 3.25. Data for the freckling propensity of a number of experimental single-crystal superalloys [18]. (a) Average number of freckle chains per blade sample; (b) corresponding differences between the densities of the unpartitioned and partitioned liquid at an interval of 20 K below the liquid of the alloy concerned. Note the correlation between the data in (a) and (b).

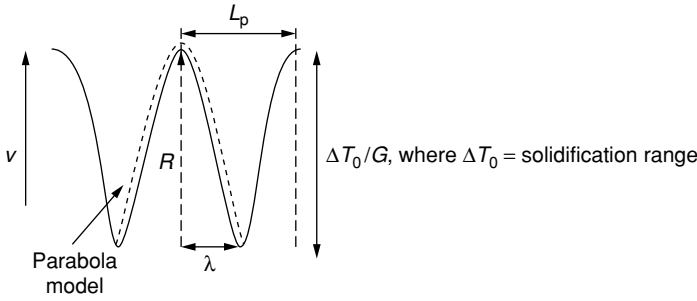


Fig. 3.26. Schematic illustration of the geometry of the dendritic structure at steady-state: the primary arm spacing is L_p , and the major and minor axes of the paraboloid of revolution fitted to the dendrite tip are $\Delta T_0/G$ and $\lambda = L_p/2$, respectively.

effect. Thus, provided that the undercooling is small,

$$\begin{aligned}\Delta T &= \Delta T_{\text{solutal}} + \Delta T_{\text{curvature}} \\ &= mc_0\Omega(1-k) + 2\theta/R\end{aligned}\quad (3.13)$$

since, for $\Omega \sim 0$, it can be demonstrated that $\Delta T_{\text{solutal}}$ is approximately $mc_0\Omega(1-k)$. The term θ is the Gibbs–Thomson coefficient; the factor of 2 arises because the dendrites are taken to be paraboloids of revolution, so that their tips are hemispherical. Consistent with the growth of an isolated dendrite tip, see the Appendix to this chapter, the supersaturation, Ω , is given by $\Omega = Rv/D$, where D is the diffusion coefficient of solute in the liquid ahead of the front. Making this substitution, along with $R = \lambda^2 G/\Delta T_0$, one has

$$\Delta T = mc_0(1-k) \left(\frac{v}{2D} \right) \left(\frac{\lambda^2 G}{\Delta T_0} \right) + \frac{2\theta \Delta T_0}{\lambda^2 G} \quad (3.14)$$

Equation (3.14) represents an expression for the dependence of the undercooling, ΔT , upon the three variables $\lambda = L_p/2$, G and v . One of these can be eliminated using a further equation or criterion [22]. For example, one can invoke the *extremum condition*, which states that growth will occur at whatever λ minimises ΔT or maximises v . This implies that $\partial \Delta T / \partial \lambda = 0$; after some algebraic manipulation, one can show that

$$vG^2\lambda^4 = 4\theta Dmc_0(1-k) \quad (3.15)$$

or alternatively

$$vG^2L_p^4 = 64\theta Dmc_0(1-k) \quad (3.16)$$

Note that the terms on the right-hand side of Equation (3.16) are expected to be invariant for any given alloy system. Thus, the primary arm spacing, L_p , is predicted to be proportional to $G^{-1/2}v^{-1/4}$. It follows that G is expected to have a more profound influence on the spacing than v – as found, for example, in the liquid metal cooling process. The experimental results

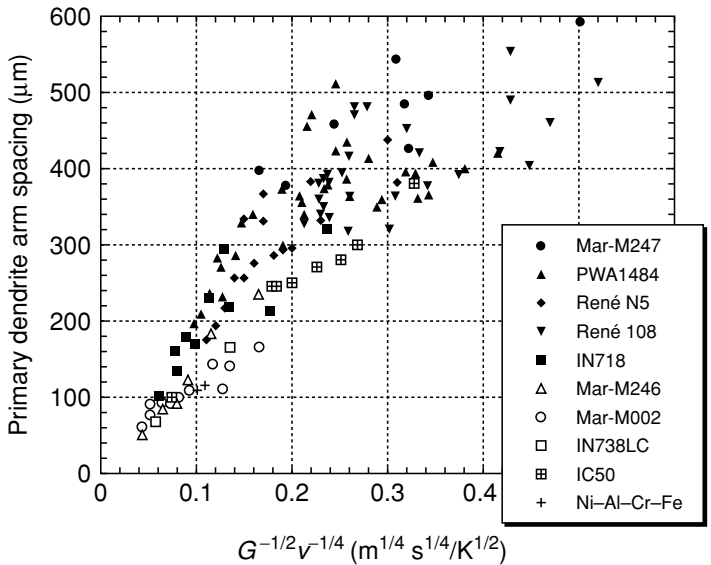


Fig. 3.27. Data [23] for the variation of the primary dendrite arm spacing with the combination $G^{-1/2}v^{-1/4}$, where G is the temperature gradient and v is the withdrawal velocity, for a number of commercial superalloys.

confirm this (see Figure 3.27), which provides data [23] for the primary dendrite arm spacing for a number of commercial superalloys solidified under a range of G and v conditions. The agreement between the experimental data and the theory gives some credence to the use of the extremum condition in deriving it, despite the lack of a firm physical basis for its application.

3.2 Optimisation of the chemistry of single-crystal superalloys

Over a period of about 20 years, the chemistry of the single-crystal superalloys has been refined in order to improve the properties displayed. Although much of this progress has been incremental, a clear picture has emerged of the roles played by the various elements in conferring the best properties. Whilst it should be remembered that the various alloy producers and engine manufacturers have settled on proprietary chemistries to maintain their competitive advantage, the compositions of the single-crystal superalloys have evolved significantly in the period since 1980 when the first ones emerged. The compositions of important first-, second-, third- and fourth-generation single-crystal superalloys are given in Tables 3.2, 3.3, 3.4 and 3.5. Of note is the introduction of significant additions of Re, and the reduction of Ti and Mo to low concentrations. Some interesting insights into the early alloy development efforts are given in refs. [24] and [25]. With the benefit of hindsight, the improvements in performance have been won by respecting the four guidelines set out on pp. 152–170.

Table 3.2. *The compositions (in weight%) of selected first-generation single-crystal superalloys*

Alloy	Cr	Co	Mo	W	Al	Ti	Ta	Nb	V	Hf	Ni	Density (g/cm ³)
Nasair 100	9	—	1	10.5	5.75	1.2	3.3	—	—	—	Bal	8.54
CMSX-2	8	4.6	0.6	8	5.6	1	6	—	—	—	Bal	8.60
CMSX-6	9.8	5	3	—	4.8	4.7	2	—	—	0.1	Bal	7.98
PWA1480	10	5	—	4	5	1.5	12	—	—	—	Bal	8.70
SRR99	8	5	—	10	5.5	2.2	3	—	—	—	Bal	8.56
RR2000	10	15	3	—	5.5	4	—	—	1	—	Bal	7.87
Rene N4	9	8	2	6	3.7	4.2	4	0.5	—	—	Bal	8.56
AM1	7.8	6.5	2	5.7	5.2	1.1	7.9	—	—	—	Bal	8.60
AM3	8	5.5	2.25	5	6	2	3.5	—	—	—	Bal	8.25
TMS-6	9.2	—	—	8.7	5.3	—	10.4	—	—	—	Bal	8.90
TMS-12	6.6	—	—	12.8	5.2	—	7.7	—	—	—	Bal	9.07

Table 3.3. *The compositions (in weight%) of selected second-generation single-crystal superalloys*

Alloy	Cr	Co	Mo	Re	W	Al	Ti	Ta	Nb	Hf	Ni	Density (g/cm ³)
CMSX-4	6.5	9	0.6	3	6	5.6	1	6.5	—	0.1	Bal	8.70
PWA1484	5	10	2	3	6	5.6	—	8.7	—	0.1	Bal	8.95
Rene N5	7	8	2	3	5	6.2	—	7	—	0.2	Bal	8.70
MC2	8	5	2	—	8	5	1.5	6	—	—	Bal	8.63
TMS-82+	4.9	7.8	1.9	2.4	8.7	5.3	0.5	6	—	0.1	Bal	8.93

Table 3.4. *The compositions (in weight%) of selected third-generation single-crystal superalloys*

Alloy	Cr	Co	Mo	Re	W	Al	Ti	Ta	Nb	Hf	Others	Ni	Density (g/cm ³)
CMSX-10	2	3	0.4	6	5	5.7	0.2	8	0.1	0.03	—	Bal	9.05
Rene N6	4.2	12.5	1.4	5.4	6	5.75	—	7.2	—	0.15	0.05C 0.004B 0.01Y	Bal	8.97
TMS-75	3	12	2	5	6	6	—	6	—	0.1	—	Bal	8.89

Table 3.5. *The compositions (in weight%) of selected fourth-generation single-crystal superalloys*

Alloy	Cr	Co	Mo	Re	Ru	W	Al	Ti	Ta	Hf	Others	Ni	Density (g/cm ³)
MC-NG	4	<0.2	1	4	4	5	6.0	0.5	5	0.10	—	Bal	8.75
MX4 /	2	16.5	2.0	5.95	3	6.0	5.55	—	8.25	0.15	0.03C	Bal	9.20
PW1497											0.004B		
TMS-138	2.8	5.8	2.9	5.1	1.9	6.1	5.8	—	5.6	0.05	—	Bal	8.95
TMS-162	2.9	5.8	3.9	4.9	6	5.8	5.8	—	5.6	0.09	—	Bal	9.04

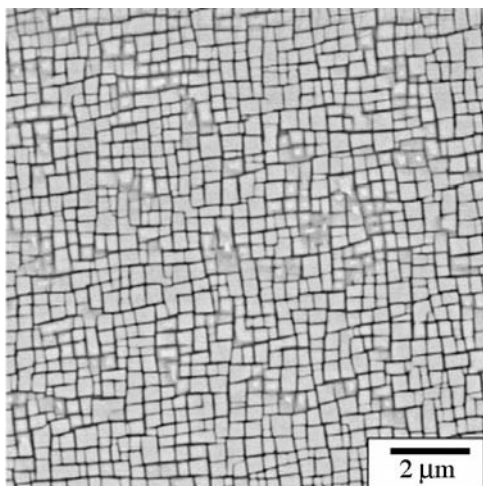


Fig. 3.28. Scanning electron micrograph of the γ/γ' microstructure of the CMSX-4 single-crystal superalloy. (Courtesy of Nirundorn Matan.)

3.2.1 Guideline 1

Proportions of γ' -forming elements such as Al, Ti and Ta should be high, such that the γ' fraction is $\sim 70\%$.

Why is 70% an appropriate figure? To answer this question, one must have an appreciation of the micromechanisms of creep deformation in the single-crystal superalloys. This is discussed in more detail in Section 3.3, but here it is sufficient to note that, across a wide range of temperature and levels of applied stress, creep deformation on the microscale is restricted to the γ channels which lie between the precipitates of γ' . Thus, creep dislocations do not penetrate the γ' precipitates. It follows then that a reduced fraction of γ yields improved properties. The optimum microstructure then consists of many fine γ' particles with very thin layers of the γ matrix separating them; see Figure 3.28. This can be rationalised by making a rough estimate of the Orowan stress, τ_{loop} , following ref. [26]. This is approximately Gb/l , where G is the shear modulus, b is the Burgers vector and l is the width of the γ' channel. With $G = 50$ GPa, $b = 0.25$ nm and $l = 100$ nm, this yields 125 MPa. For uniaxial loading along $\langle 100 \rangle$, the Schmid factor assuming $\{111\}\langle 1\bar{1}0 \rangle$ slip is $1/\sqrt{6}$; the applied stress for bowing is then 306 MPa. This is significant, and is one of the major reasons why the single-crystal superalloys perform so well at elevated temperatures.

A second point is that the creep performance does not increase monotonically as the γ' fraction is increased – thus, an alloy with 70% γ' performs better than one with 100% γ' . This implies a strong strengthening effect from the γ/γ' interfaces, which impart resistance to creep deformation. This effect has been confirmed by measuring the creep rupture lives of a series of alloys containing various fractions of γ' of identical composition [27]. The set of tie-lines characterising a multicomponent alloy of interest was first determined, and a family of related alloys cast with chemistries chosen such that the compositions of both γ and γ' were identical in all cases. The results are given in Figure 3.29. Note that the

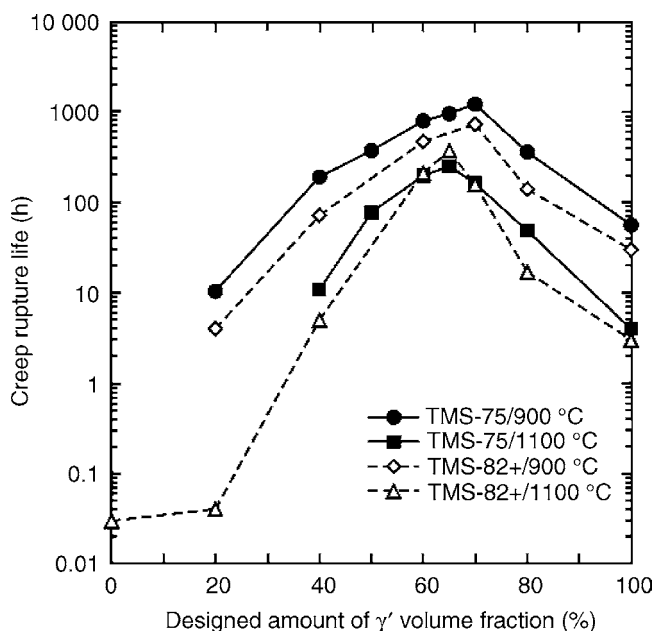


Fig. 3.29. Variation of the creep rupture lives of the single-crystal superalloys TMS-75 and TMS-82+, as a function of the amount of γ' phase [27]. The creep rupture life is largest when the γ' fraction is about 70%. (Courtesy of Hiroshi Harada.)

creep strength of γ' is somewhat superior to that of γ , but that microstructures consisting of mixtures of γ and γ' perform best. The optimum γ' fraction is $\sim 70\%$.

With respect to the optimisation of the chemistry, it is useful to consider the Ni–Al binary diagram. At 700 °C, the solubility of Al in Ni is about 10 at%. The Ni_3Al has an Al content of 25 at%. Therefore, in the binary system one needs a composition, $\bar{c}$, satisfying $(\bar{c} - 10)/(25 - 10) = 0.75$. Hence $\bar{c} = 21.25$ at%. This corresponds to about 11 wt%. A modern single-crystal superalloy would have about 6 wt% Al. Clearly, significant reliance is being placed on the γ' -forming elements such as Ta and Ti.

Thus the balance of γ - and γ' -forming elements must be chosen such that the γ' fraction is optimal. But there is another consideration. There should exist a sufficient solutioning window, ΔT , below the melting temperature, across which γ is the only stable phase. This is necessary because, after solidification, heat treatment is required to remove residual microsegregation and eutectic mixtures rich in γ' . This puts the γ' precipitates into solution; during subsequent exposure to lower temperatures, the γ' develops a uniform microstructure with an optimum precipitate size. The heat treatment for a typical single-crystal superalloy such as CMSX-4 would involve a solutioning of 8 h at 1314 °C, a primary age of 4 h at 1140 °C followed by a secondary age of 16 h at 870 °C. Figure 3.30 shows a prediction of the phase stability of the CMSX-4 superalloy around the melting temperature. The solutioning window is about 50 °C, which is rather small; this emphasises the need for precise temperature control during heat treatment. Since the solutioning window is expected to decrease as the γ' fraction is increased, the heat treatment of the higher-strength alloys

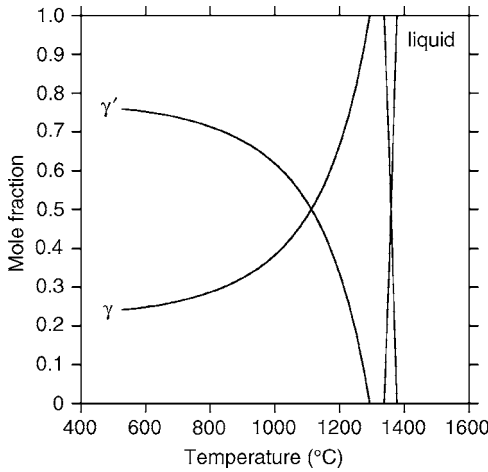


Fig. 3.30. Variation of the fraction of the liquid, γ and γ' phases with temperature for the single-crystal superalloy CMSX-4. Note the solvating window of about 50 °C which lies between 1300 °C and 1350 °C.

can be difficult to accomplish in practice. For the third- and fourth-generation alloys which are rich in Re, solvating times are substantially longer.

3.2.2 Guideline 2

The composition of the alloy must be chosen such that the γ/γ' lattice misfit is small; this minimises the γ/γ' interfacial energy so that γ' coarsening is restricted.

The lattice misfit, δ , is defined according to

$$\delta = 2 \times \left[\frac{a_{\gamma'} - a_{\gamma}}{a_{\gamma'} + a_{\gamma}} \right] \quad (3.17)$$

where the lattice parameters of the γ and γ' phases, a_{γ} and $a_{\gamma'}$, respectively, depend via Vegard's law on the elemental compositions in each phase, and consequently on the extent of γ/γ' partitioning. For the γ phase, one has, to a good approximation (refer to Figure 2.18),

$$a_{\gamma} = 3.524 + 0.110x_{Cr}^{\gamma} + 0.478x_{Mo}^{\gamma} + 0.444x_{W}^{\gamma} + 0.441x_{Re}^{\gamma} \\ + 0.179x_{Al}^{\gamma} + 0.422x_{Ti}^{\gamma} + 0.700x_{Ta}^{\gamma} \text{ \AA} \quad (3.18)$$

where the x_i^{γ} are the mole fractions of i in γ , and the various constants are taken from correlations such as those given in Chapter 2. For the γ' phase, one finds

$$a_{\gamma'} = 3.570 - 0.004x_{Cr}^{\gamma'} + 0.208x_{Mo}^{\gamma'} + 0.194x_{W}^{\gamma'} + 0.262x_{Re}^{\gamma'} + 0.258x_{Ti}^{\gamma'} + 0.500x_{Ta}^{\gamma'} \text{ \AA} \quad (3.18b)$$

where the mole fractions are now evaluated in that phase. The value of the misfit has been found to exhibit a significant effect on the morphology of the γ' precipitates [28]; see Figure 3.31. When the misfit is small, less than about 0.5%, the γ' particles are cuboids

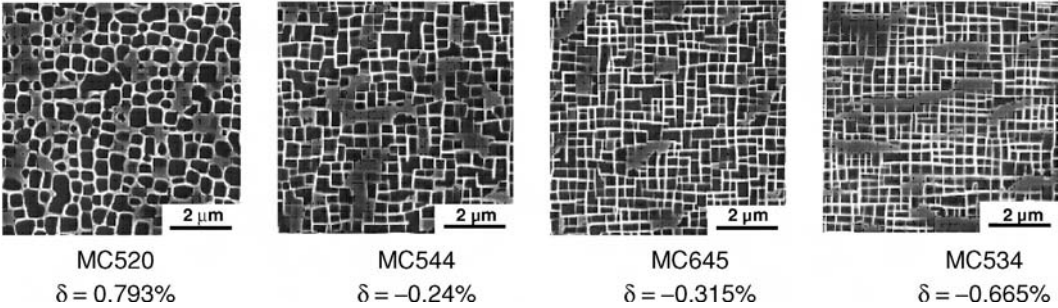


Fig. 3.31. The γ/γ' microstructures displayed by four experimental single-crystal superalloys of varying chemistries [28]. Note the correlation between the γ' morphology and the estimated values of the lattice misfit, δ . (Courtesy of Pierre Caron.)

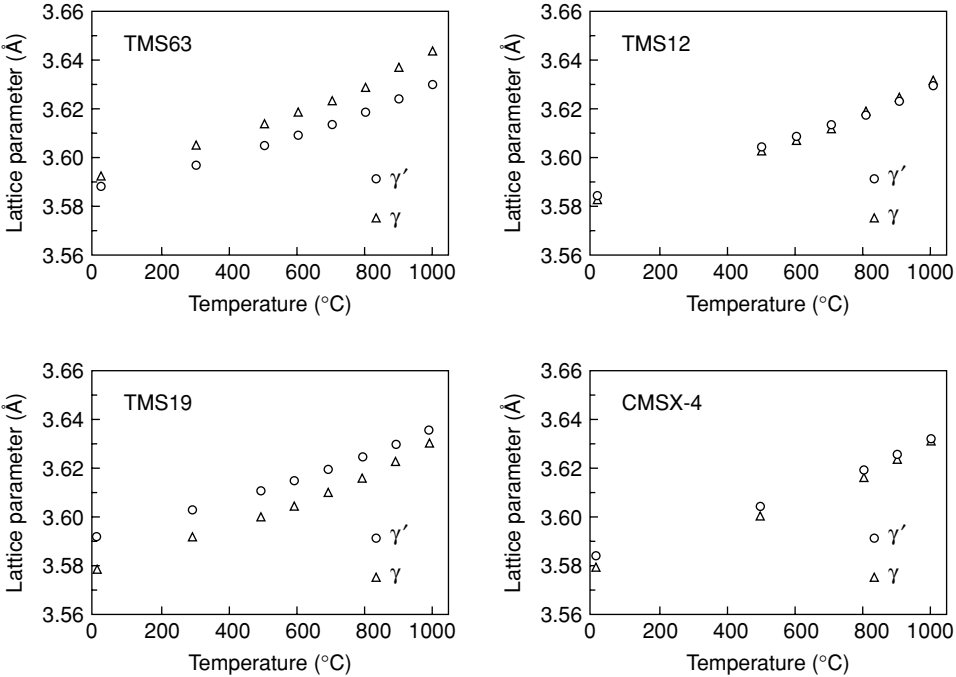


Fig. 3.32. Variation of the lattice parameter of the γ matrix and γ' precipitates in a number of nickel-based single-crystal superalloys, determined using X-ray diffraction. Data from ref. [29]. (Courtesy of Hiroshi Harada.)

with sharp corners characteristic of elastic coherency. The precipitates become spherical as the magnitude of the misfit increases, since coherency is lost.

When designing alloys, an additional consideration is the strong temperature dependence of the lattice misfit. Experiments using high-temperature diffractometry indicate that the expansion coefficient of γ' is considerably less than that of γ ; thus, for all known single-crystal superalloys the lattice misfit becomes *more negative* as the temperature increases. This is demonstrated in Figure 3.32, in which results are given for a_γ and $a_{\gamma'}$ for four alloys

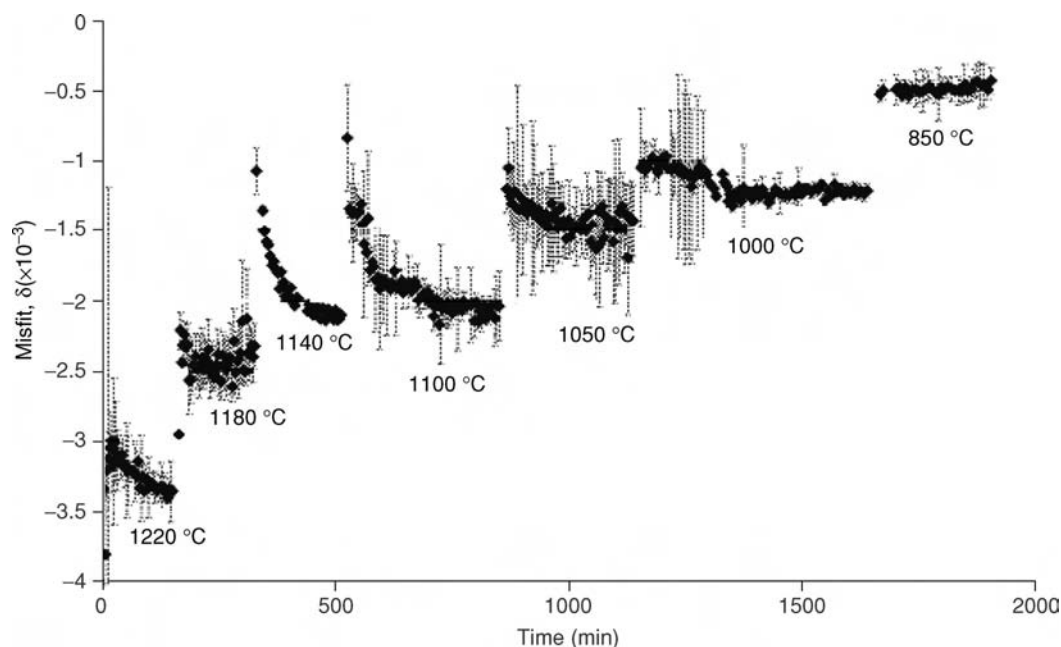


Fig. 3.33. Measurements of the lattice misfit, δ , in an experimental single-crystal superalloy at various temperatures between 1220 °C and 850 °C [30]. Note the finite time at each temperature required for equilibrium to become established. (Courtesy of Giovanni Bruno.)

of differing compositions [29]. Note that in some cases $a_{\gamma'} > a_{\gamma}$, but for others $a_{\gamma'} < a_{\gamma}$, implying that the misfit, δ , can be either positive or negative. In Figure 3.33, the variation of the lattice misfit of an experimental superalloy during cooling from 1220 °C to 850 °C is given [30], for which there were several isothermal holds. It is significant that after each step change in temperature, a finite time is required for δ to attain its equilibrium value. This is associated with elemental partitioning between the γ and γ' phases, and the establishment of a coherent γ/γ' interface and the equilibration of coherency stresses.

A consequence of a non-zero δ , or alternatively a zero δ followed by a change in temperature, are *misfit stresses* on the scale of the γ/γ' microstructure, which are present even when the externally applied load is zero. Consider the SRR99 single-crystal superalloy. It has been reported [31] that $a_{\gamma'} = 0.35837$ nm and $a_{\gamma} = 0.35887$ nm at room temperature, which yields a negative misfit, δ , of -1.4×10^{-3} . Assuming that the γ/γ' interfaces are forced into elastic coherence, it follows that the γ' particles are constrained in a state of hydrostatic tension, and that the γ channels separating them are in a state of biaxial compression; see Figure 3.34. The effect of elastic coherency is thus to *increase* $a_{\gamma'}$ from its value that would be measured if the particles are removed, which is possible by means of wet chemical electrolysis; conversely, a_{γ} is decreased. One should then refer to δ as the *constrained* lattice misfit to distinguish it from an *unconstrained* value, which is expected to be smaller in magnitude.

In practice, it has proven possible to design alloys with a constrained lattice misfit, δ , measured at room temperature, within $\pm 1\%$. Alloys such as CMSX-4, Rene N5 and Rene

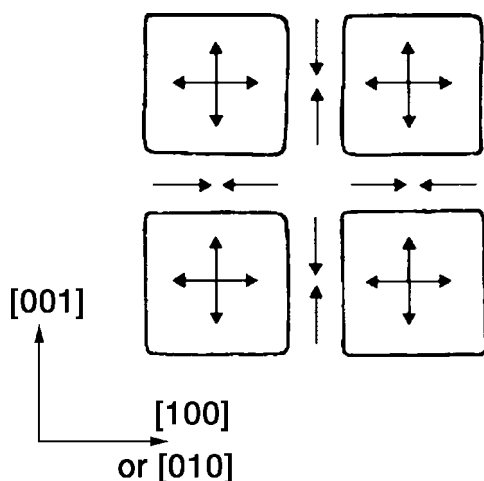


Fig. 3.34. Schematic illustration of the coherency stresses to be expected in a nickel-based superalloy when the misfit, δ , is negative. (Courtesy of Hael Mughrabi.)

N6 have values in the range 0 to +0.4%. There is some evidence, see Figure 3.35, that alloys of negative misfit possess greater creep resistance than those with positive misfit [29]. This is because creep deformation is controlled by dislocation activity in the γ channels promoted by externally applied tensile stresses, which are superimposed on the misfit stresses. As a consideration of Figure 3.34 confirms, the net stresses seen in the γ channels will be lower in a negatively misfitting alloy than in a positively misfitting one, for loading along the normal to the unit cell faces, $\langle 001 \rangle$. The influence of the lattice misfit on the creep performance is discussed further in Section 3.3.

3.2.3 Guideline 3

Concentrations of creep-strengthening elements, particularly Re, W, Ta, Mo and Ru, must be significant but not so great that precipitation of topologically close-packed (TCP) phases is promoted.

The success of this strategy can be judged by considering the creep performance of the first-, second- and third-generation single-crystal superalloys. The data in Figure 1.16 indicate that creep rupture lives vary from about 250 h at 850 °C/500 MPa for a first-generation alloy such as SRR99, to about 2500 h for a third-generation alloy such as CMSX-10/RR3000. Under more demanding conditions, for example 1050 °C/150 MPa, rupture life improves four-fold from 250 h to 1000 h. As a consideration of Table 1.3 shows, the major differences between the chemistries of these first-, second- and third-generation alloys lies in the addition of Re. Although absent in the first-generation alloys, the second-generation alloys contain about 3 wt%Re, and the third-generation alloys contain between 5 and 6 wt%. Thus, a strong creep-strengthening effect can be attributed to Re. The elements W and Ta appear to be less potent, with Cr and Co having little effect. Experience has shown that improvements

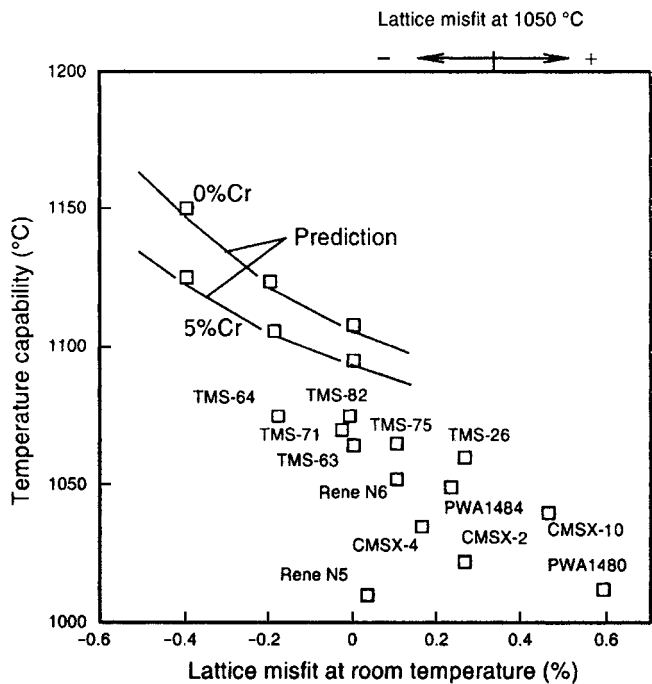


Fig. 3.35. Variation with lattice misfit of the creep capability, characterised in terms of the temperature for 1000 h creep life at a stress of 137 MPa. (Courtesy of Hiroshi Harada.)

in creep strengthening occur in the following order:



The best strengtheners thus appear to be those which diffuse the slowest in nickel – this is unsurprising, given the importance of diffusion in the dislocation climb processes, when mass transport at dislocation cores is required.

In practice, there is a limit to the concentrations of the refractory elements which can be added. Excessive quantities of Cr, Mo, W and Re promote the precipitation of intermetallic phases which are rich in these elements, as discussed in Section 2.1.3. Precipitation of these TCP phases implies a lack of stability, which is detrimental to creep behaviour. Examine the Larson–Miller diagram given in Figure 3.36, on which the creep performance of a number of prototype second-generation alloys containing 3 wt%Re is plotted. At low temperatures of about 800 °C and stresses of 500 MPa, the behaviour is found to vary only weakly with alloy composition. However, at higher temperatures of ~1000 °C and stresses of ~150 MPa, the alloys fall into two groups of contrasting behaviour. Creep performance is very much inferior when TCP formation occurs, with the rupture life being reduced accordingly. Figure 3.37 illustrates the type of extensive TCP formation characteristic of the unstable alloys. The conclusion that can be drawn from these experiments is that the TCP precipitation is detrimental to properties. The depletion of the remaining γ/γ' structure of the elements which improve creep strength is responsible for this effect.

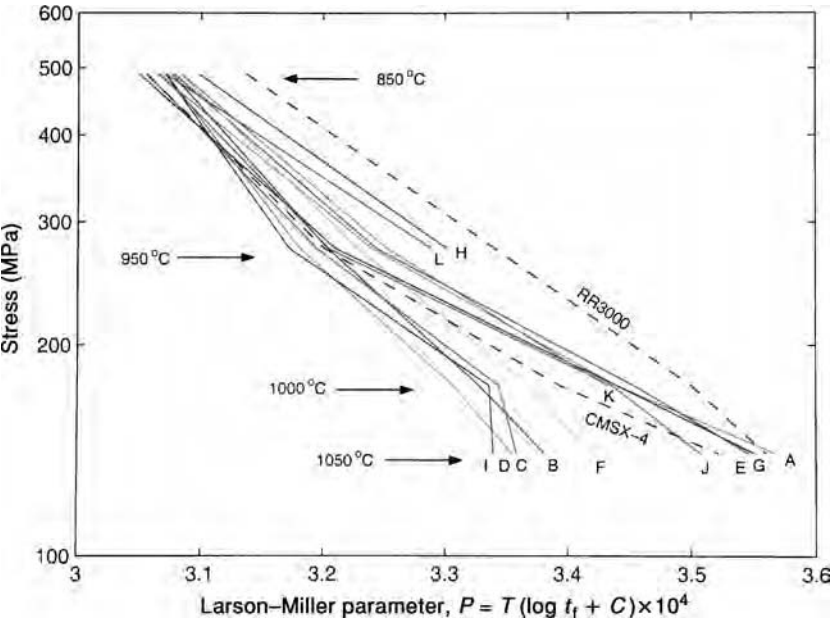


Fig. 3.36. Variation of the stress for rupture with the Larson–Miller parameter, P , for various second-generation single-crystal superalloys. The alloys B, C, D, F and I were found to be prone to TCP formation; thus the creep properties of the stable alloys are very much superior. (Courtesy of Martin Rist.)

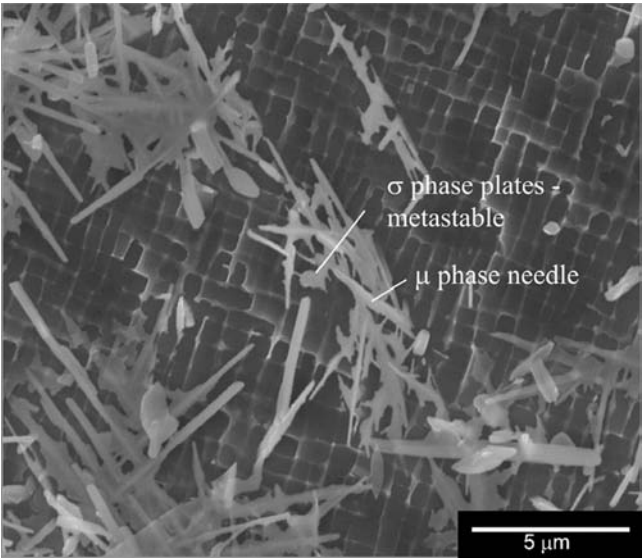


Fig. 3.37. Scanning electron micrograph of the experimental second-generation superalloy RR2071, aged at 900 °C for 500 h, showing extensive precipitation of the TCP phases σ and P . (Courtesy of Cathie Rae.)

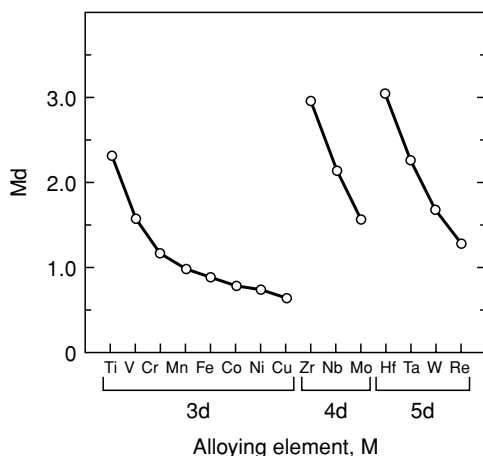


Fig. 3.38. The values of the Md parameters for some of the transition metals [32]; note the strong correlation with atomic number.

Because of the profound influence of TCP precipitation on creep behaviour, efforts have been made to develop predictive approaches which enable alloy developers to decide whether any given alloy composition is unstable or not. In view of the nature of the phases, band structure calculations are the most appropriate. For example, for the first row of transition metals, the 4p fraction of the hybrids can be ignored and the outer electrons regarded as existing in two distinct, overlapping bands derived from the 3d band (providing up to ten electrons) and the 4s band (providing two). Similar arguments apply to the second and third rows. These considerations have allowed the energy levels (in eV) to be determined using band theory for the transition metals at, or around, the Fermi level. These have been termed metal-d levels (symbol Md), and have been found to correlate well with the electronegativity and the metallic radius. In the absence of more rigorous electron theory, one then works out the effective $\overline{\text{Md}}_{\text{eff}}$ level from [32]

$$\overline{\text{Md}}_{\text{eff}} = \sum_i \bar{x}_i \text{Md}_i \quad (3.19)$$

where the values of Md_i for the various i values are given in Figure 3.38, and the $\bar{x}_i$ correspond to the mole fraction of i in the alloy. The *higher* value of $\overline{\text{Md}}_{\text{eff}}$ the higher the probability of TCP formation. The critical value appears to around 0.99. The procedure described here is a simplified version of the procedure known as PHACOMP, which is an acronym for ‘phase computation’. In a variant of the procedure described here, the concentration terms in the summation can be taken to be the mole fractions in γ , since the TCPs usually precipitate from this phase. However, there is no strong evidence that this provides a greater predictive capability.

In recent years, the stability of the single-crystal superalloys with respect to TCP precipitation has been found to be strongly improved by additions of Ru. See, for example, Figure 3.39, in which the time–temperature–transformation (TTT) curves for the formation of TCP phases is given for two experimental alloys, the second of which is doped with 2 wt%Ru [33]. A consequence of this is that the creep properties are found to be significantly improved (see Figure 3.40), particularly at temperatures beyond about 950 °C;

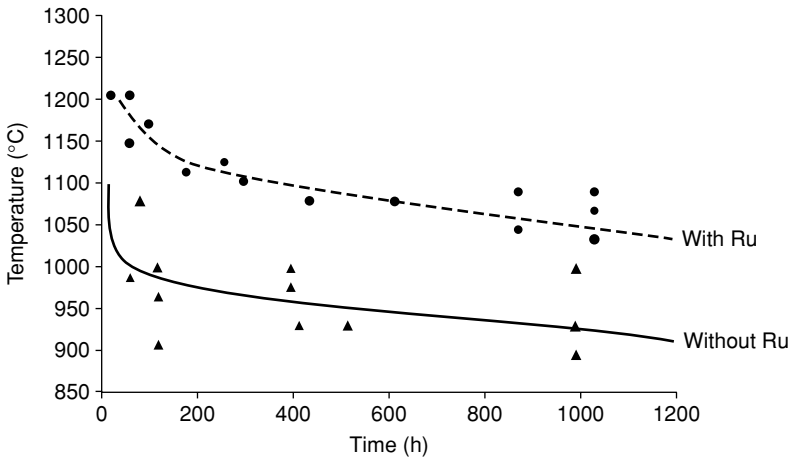


Fig. 3.39. Time–temperature–transformation (TTT) diagrams for the formation of TCP phases in two single-crystal superalloys, one with and the other without Ru-doping – the base composition being Ni–12Co–2.5Cr–9W–6.4Re–6Al–5.5Ta–0.15Hf (wt%). (Courtesy of An-Chou Yeh.)

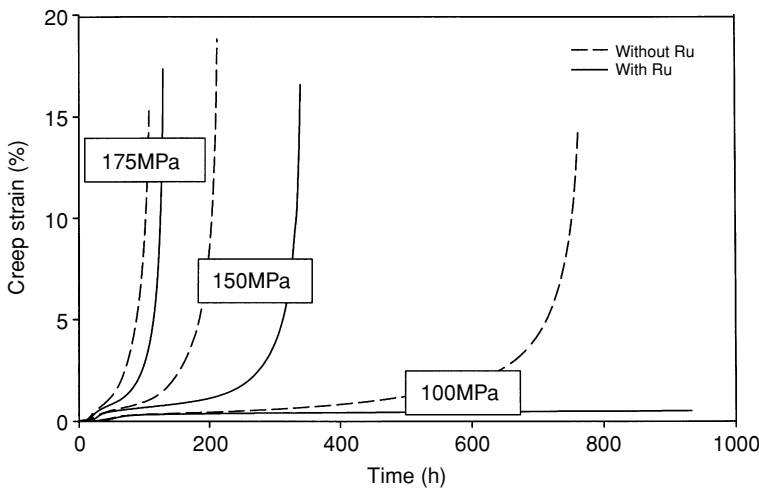


Fig. 3.40. Creep curves for two single-crystal superalloys tested at 1100 °C, the second doped with 2 wt%Ru [33] – the base composition being Ni–12Co–2.5Cr–9W–6.4Re–6Al–5.5Ta–0.15Hf (wt%). (Courtesy of An-Chou Yeh.)

for this reason, Ru additions are now commonly added to the latest ‘fourth-generation’ single-crystal superalloys, although, at the time of writing, these are not yet being used in commercial engines. The reasons for the so-called ‘Ru effect’ are controversial; some researchers have noted that Ru, which partitions strongly to the γ phase, has a strong influence on the γ/γ' partitioning behaviour of elements such as Al, Re and Cr [34,35]. Significant additions of Ru have the effect of reducing the partitioning coefficients of many of the alloying additions (see Figure 3.41); this effect, however, is not unique to Ru, since other

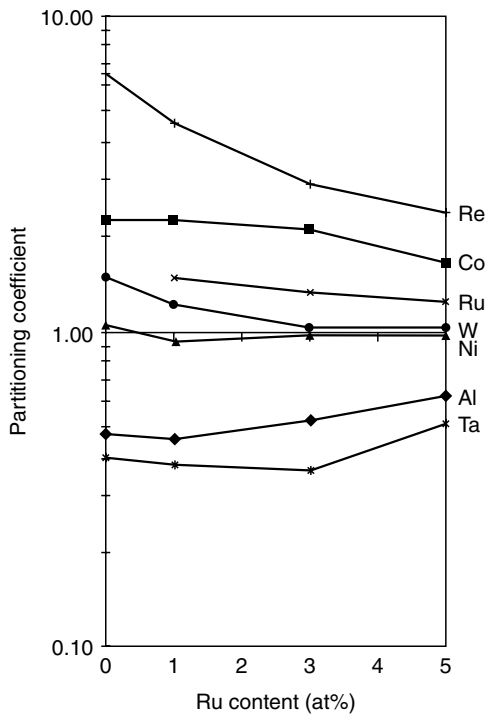


Fig. 3.41. Measurements of the influence of Ru-doping on the γ/γ' partitioning coefficient of a number of elements in a model single-crystal superalloy system [35]. (Courtesy of Tony Ofori.)

platinum-group metals have a similar effect. Thus, whilst Ru does undoubtedly influence partitioning behaviour, it is not established that any so-called ‘reverse partitioning’ is the reason for Ru’s beneficial effect on the alloy stability. For example, one cannot rule out at this stage the possibility that Ru poisons the nucleation kinetics or else retards the growth rates of the TCP phases.

3.2.4 **Guideline 4**

The composition must be chosen such that surface degradation through exposure to the hot, working gases is avoided.

Attack by oxidation is a possibility. When this occurs, wall thickness and thus loading bearing capacity are lost, but more importantly it can be the precursor to mechanical failure by fatigue. One should examine first the thermodynamics of formation of the important oxides NiO, Cr₂O₃ and Al₂O₃. The Gibbs energies of formation indicate that Al₂O₃ is the most stable of the three, NiO the least, with Cr₂O₃ between the two – as demonstrated by a consideration of the Ellingham diagram. On this basis alone, one might expect Al₂O₃ to be the oxide of prevalence in superalloy systems. However, of equal relevance in practice are the kinetics of oxidation. NiO has a high concentration of nickel vacancies and thus forms very rapidly with voids and microcracking prevalent – thus it is friable, which promotes spallation and hence sustained attack. Chromia, Cr₂O₃, is less so, but at elevated temperatures it is

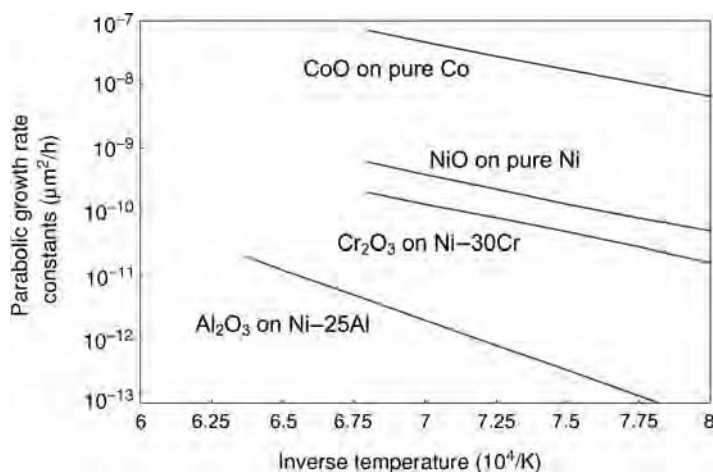


Fig. 3.42. Variation of the parabolic thickening constants for oxide formation with temperature, for various species on different substrates. Note that the formation of Al_2O_3 is slow.

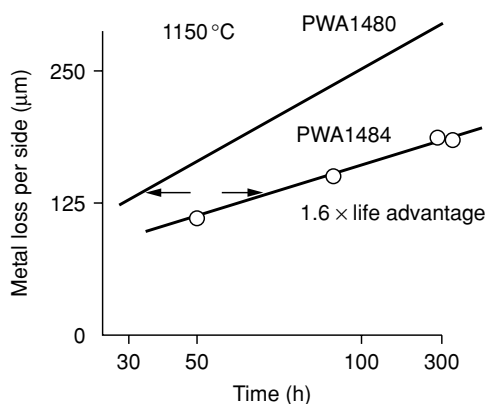


Fig. 3.43. Data for the rate of metal loss due to oxidation of the PWA1480 and PWA1484 single-crystal superalloys, confirming the improved oxidation behaviour of the second-generation alloy. Data taken from ref. [36].

oxidised to the gaseous CrO_3 , with a significant rate of oxidation being determined by transport through a gaseous boundary layer, which is greatly increased in rapidly moving gas streams. This effect limits the temperature at which a chromia-forming alloy can be used to about 900°C . In binary Ni–Cr alloys, Cr_2O_3 will not form if the Cr content falls below about $\sim 12\text{ wt}\%$, although alloying with Al reduces this value. When growth of an Al_2O_3 -based scale occurs, the rate of oxidation is always found to be slow. Typical parabolic rate constants for the various oxides are given in Figure 3.42. For these reasons, the first single-crystal superalloys – for example the first- and second-generation ones – were designed to be alumina formers. To ensure this, the aluminium content was chosen to be around $\sim 6\text{ wt}\%$.

It is instructive to consider the improvements in oxidation resistance which arose as the second-generation single-crystal alloys were developed. Consider Pratt & Whitney's success with PWA1484, which is a derivative of the first-generation alloy PWA1480 [36]. Figure 3.43, which compares the oxidation performance of the two alloys during isothermal

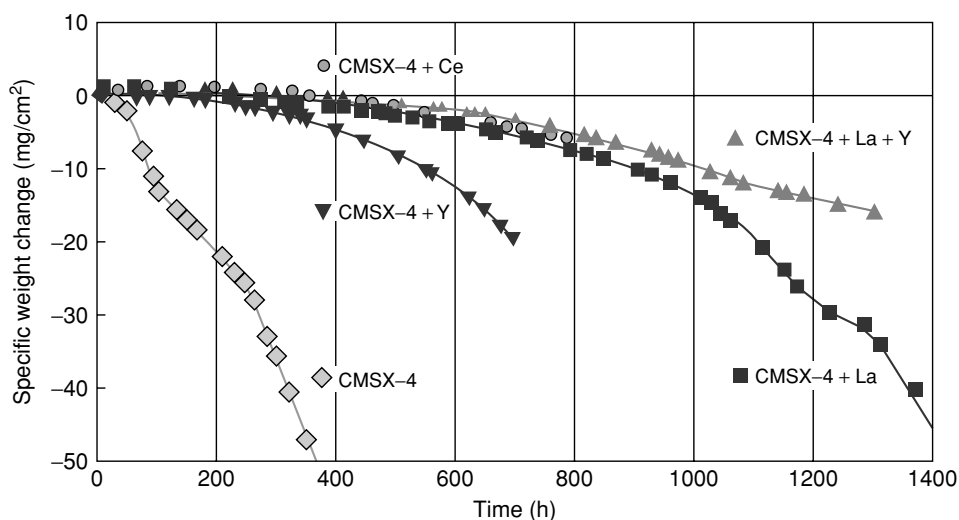


Fig. 3.44. Weight change data [38] due to cyclic oxidation for the CMSX-4 single-crystal superalloy during a cyclic oxidation test with a peak temperature of 1093 °C, and for variants of it doped with the rare-earth elements La, Ce and Y. (Courtesy of Jackie Wahl.)

exposure to 1000 °C, indicates that the rate of metal loss due to oxidation is very much less for PWA1484, with an improvement of $\times 1.6$ in life being claimed. The reasons for this are as follows. The Al content was increased from 5.0 to 5.6 wt%, to increase the tendency for Al_2O_3 -scale formation. Secondly, the Ti content was reduced to nearly zero, since it was found that Ti increases the rate of alumina formation due to its solubility in it; since the Ti^{4+} and Al^{3+} ions have different charges, vacancies are introduced to the alumina lattice, thus increasing the ionic mobilities. Finally, additions of rare-earth additions, such as Hf, La and Y, were found to be beneficial – this is because they bind strongly with S, which is inevitably present at impurity levels – the so-called ‘gettering effect’. It is now known that S present in the superalloys has a strong influence on the oxidation resistance – due to its solubility in the oxide scale and its likely segregation to the metal/scale interface – it has been suggested, for example, that sulphur destroys the strong Van der Waals bond between the alumina scale and the metal substrate [37]. This explains the considerable emphasis placed by the alloy manufacturers on reducing the S levels of superalloy melts; sulphur should be restricted to 10–20 ppm, although recent melt desulphurisation techniques have allowed residual levels to reach less than 1 ppm on a routine basis [38]. Results from dynamic cyclic oxidation tests for bare CMSX-4 with and without rare-earth doping (see Figure 3.44) provide very convincing evidence that the performance of the second-generation alloy CMSX-4 is improved by additions of the rare-earth elements. Recent experiments on the TMS-138 superalloy doped with 4, 40, 100 and 1000 ppm by weight of yttrium indicate that the oxidation resistance is optimal at about 100 ppm; field emission electron microprobe analysis (FE-EPMA) confirms that sulphur is present primarily as the YS compound at this yttrium concentration, but as Y_2S_3 when it falls to 40 ppm; see Figure 3.45. Moreover, sulphur is present in the oxide scale for the lower yttrium contents considered. For the very

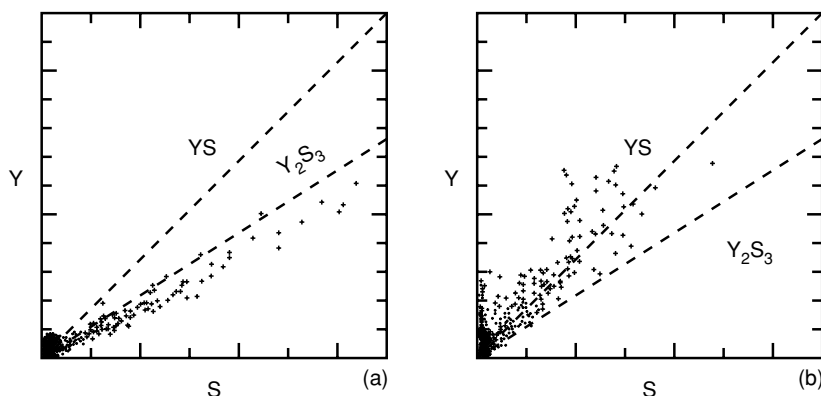


Fig. 3.45. Field emission electron microprobe analysis (FE-EPMA) data from secondary particles in the TMS-138 superalloy doped with (a) 40 and (b) 100 ppm (by weight) of yttrium, confirming the presence of Y_2S_3 and YS, respectively. (Courtesy of Atsushi Sato.)

best oxidation performance in single-crystal superalloys, it also appears to be necessary to tie up residual carbon, which is often present at the 400 ppm level; for this reason, carbide formers such as Hf are often added at greater than 400 ppm concentrations.

In practice, the oxidation behaviour of the single-crystal superalloys is complicated by the many different elements present. Consider, for example, a third-generation alloy such as CMSX-10, which does not form an alumina scale [39]; see Figure 3.46. During oxidation, an external scale of NiO forms very rapidly and spallation occurs readily. The underlying metal, being depleted in Ni, forms β -NiAl, which grows into the metal by a process of internal oxidation. Degradation is further complicated by the formation of various Cr-, W- and Ta-rich oxides beneath the NiO scale [39]. These effects are due to the choice of alloy composition: the Re content is high at 6 wt% to confer creep strengthening with the Cr content reduced to 2 wt% in an attempt to prevent TCP formation. The Cr content is too low to prevent either the formation of a thick external Ni scale or to resist attack by internal oxidation. For these reasons, for adequate environmental performance it is necessary to coat CMSX-10 by the process of pack or gas-phase aluminisation. A recent comparative study of the cyclic oxidation performance of typical first-, second-, third- and fourth-generation single-crystal superalloys has confirmed the superior performance of the earlier variants; see Figure 3.47. These considerations indicate that, during the development of the latest generations of single-crystal superalloys, less emphasis has been given to imparting oxidation resistance by appropriate choice of alloy chemistry; instead, one is increasingly reliant on coating technologies for imparting resistance to environmental degradation. Chapter 5 deals with these technologies in detail.

Case study – hot corrosion

Attack by hot corrosion is also a likelihood which should be accounted for when designing single-crystal superalloys [40,41]. Whilst a number of different corrosion degradation reactions can arise, the pervasive feature of them is chemical attack by gaseous, oxidising

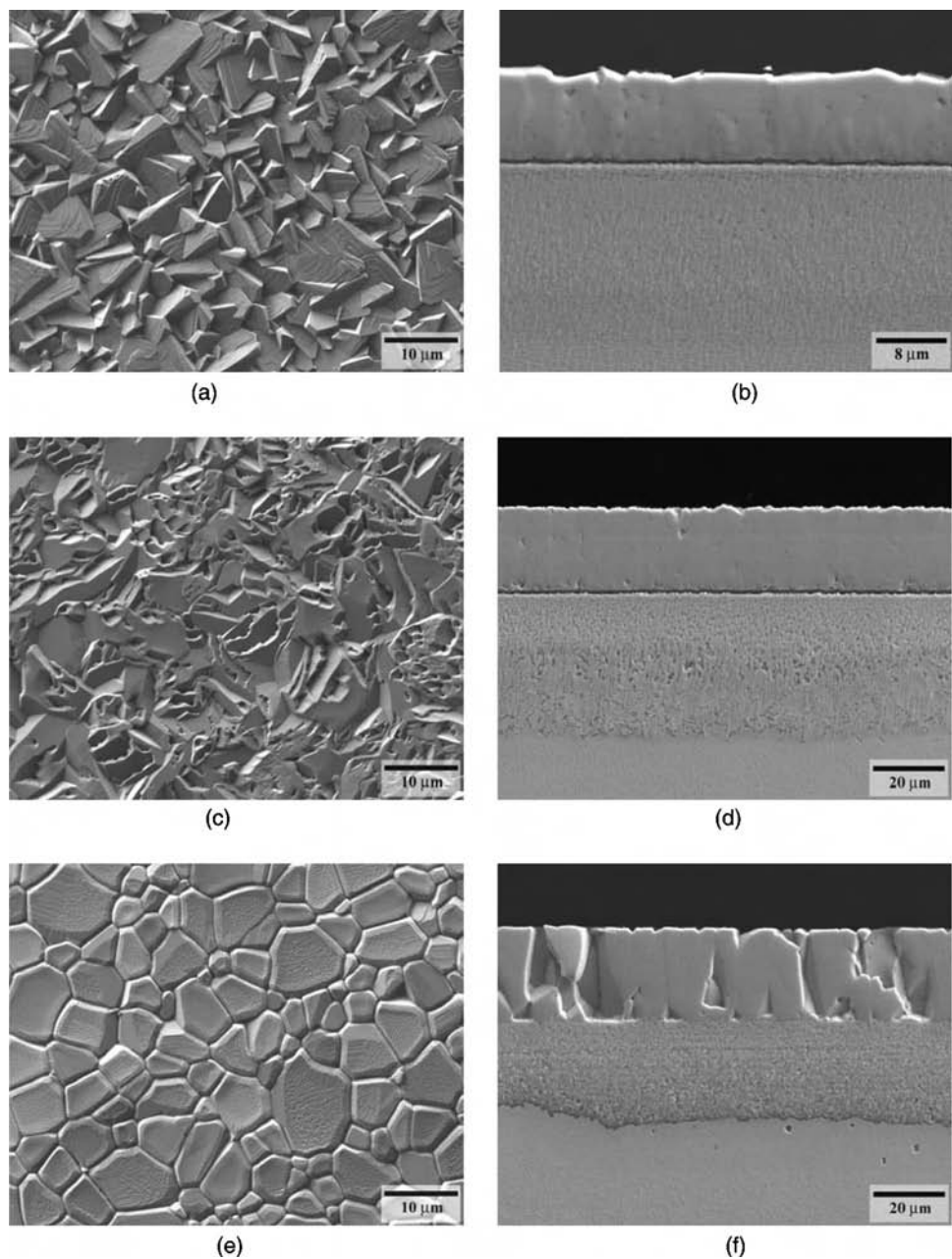


Fig. 3.46. External NiO scale and internal oxidation in the third-generation superalloy CMSX-10 following isothermal heat treatment for 100 h at (a), (b) 800 °C, (c), (d) 900 °C and (e), (f) 1000 °C, viewed from above and in cross-section. (Courtesy of Matthew Hook.)

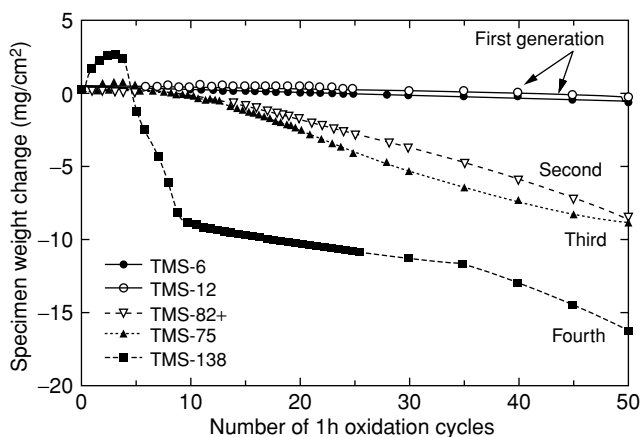
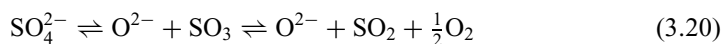


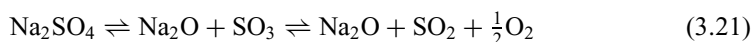
Fig. 3.47. Results from cyclic oxidation testing at 1100 °C (1 h hold at temperature) for various single-crystal superalloys. Note the superior performance of the earlier generations. (Courtesy of Kyoko Kawagishi.)

environments rich in sulphur and consequently SO_3 – the stable oxide at typical TET temperatures [42]. The sulphur is introduced since it is present as an impurity in the fuel burned in the combustor. Thus, sulphidisation – which causes the formation of nickel, chromium, cobalt and other metal sulphides, and which is thus a direct analogue of oxidation – is a possibility. However, in practice, the corrosiveness of the environment is exacerbated by its contamination with compounds (most notably NaCl, but also KCl, MgCl_2 and CaCl_2) ingested from the incoming air, which occurs particularly in marine environments; in the presence of sulphur- and oxygen-rich gases, these are converted to salts such as Na_2SO_4 , which are then deposited on the blading. Attack by deposits produced in this way is known as hot corrosion. The combination of NaCl and Na_2SO_4 is particularly pernicious, as a eutectic mixture forms when the two compounds are mixed in equimolar proportions which melts at $\sim 620^\circ\text{C}$, which is considerably lower than the melting temperatures of the pure constituents (800°C and 890°C for NaCl and Na_2SO_4 , respectively). Molten salts so produced can dissolve oxides such as Al_2O_3 and Cr_2O_3 , and thus are able to destroy any protective scale present by a fluxing mechanism. The hot-corrosion effect is exacerbated by the presence of other metallic salts containing Li, Mg, Ca or K as airborne matter and/or other impurities in the fuel such as V or Pb. For this reason, most turbine manufacturers place tolerances on the levels of fuel contaminants; see Table 3.6.

The reaction conditions characteristic of hot-corrosion attack may be either basic or acidic in nature. Basic conditions arise because of reactions of the type



or, if the prevailing ion is sodium,



Hence, the liquid becomes more basic with respect to its initial composition on account of an increase in the oxide ion concentration (the equilibria between SO_4^{2-} , O^{2-} and SO_3

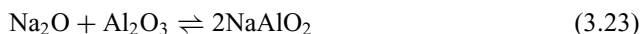
Table 3.6. *Maximum permitted fuel contaminant levels for a typical industrial gas turbine, in parts per million by weight*

Contaminant	Level
Na+K	1
V	0.5
Pb	1
F ₂	1
Cl ₂	1500
S	10 000

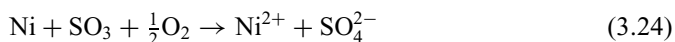
being an indicator of the acidity/basicity of the conditions). As this occurs, protective oxide scales may react with the deposit via a fluxing process according to



or, alternatively,



with the species formed being soluble in the liquid deposit. Such basic fluxing reactions have a number of distinctive features. First, this form of attack is usually restricted to high temperatures (above 900 °C) since the chemical reactions which produce oxide ions are otherwise kinetically constrained – thus it has become known as ‘high-temperature’ or ‘type I’ hot corrosion [41]. Second, a porous outer oxide scale is produced with internal sulphides in the alloy substrate near the scale–alloy interface, as a consequence of sulphur removal from the Na₂SO₄ to produce oxide ions. Finally, the amount of attack depends upon the production of oxide ions by the melt; hence, a supply of Na₂SO₄ is necessary for the attack to continue. Alternatively, acidic conditions might prevail. For example, ‘gas-phase-induced’ acidic fluxing processes can occur, for example



such that the acidic component is supplied to the liquid deposit by the gaseous atmosphere (for example SO₃ or, alternatively, V₂O₅), with the removal of nickel from the alloy preventing the development of a continuous, protective layer of either Al₂O₃ or Cr₂O₃ over the alloy surface. In this case, the attack – usually giving heavy pitting but no internal sulphides – is known as ‘low-temperature’ or ‘type II’ hot corrosion [41] since it occurs between 650 °C and 800 °C. The low-temperature restriction results from the necessity to form sulphates such as CoSO₄, Ni₂SO₄ or Al₂(SO₄)₃, which require prohibitively high SO₃ partial pressures, which are not available in typical combustion environments as the temperature is increased. So-called ‘alloy-induced’ acidic fluxing modes have also been proposed, in which the deposit becomes more acidic by reacting with certain oxides in the corrosion product, for example



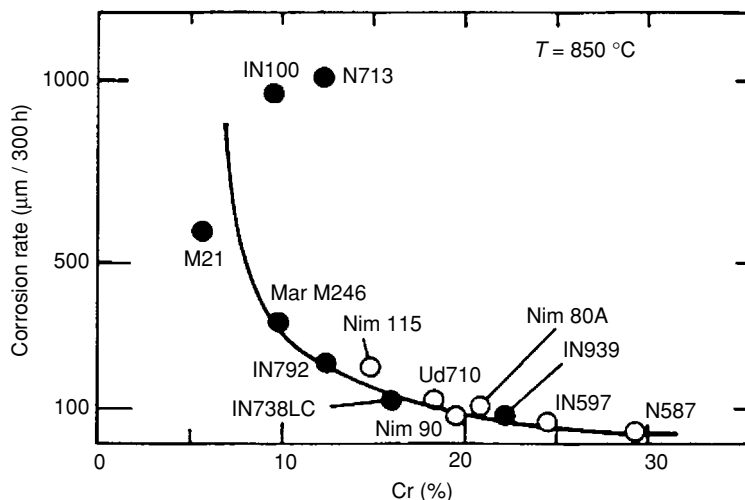


Fig. 3.48. Variation of the corrosion rate of various superalloys with Cr content, confirming the significant attack in Cr-lean systems. (Courtesy of Malcolm McLean.)

and then



implying an equilibration with SO_2 gas; subsequently reactions occur between the protective oxide barriers and the deposit according to



Alloys containing high concentrations of the refractory elements Mo, W and V are prone to this effect since they form oxides which cause Na_2SO_4 to become more acidic, with only one application of the salt being catastrophic on account of the self-sustaining nature of the reaction.

In view of such accelerated attack, which can rapidly destroy rows of turbine blades should conditions prove favourable to it, attempts have been made to design superalloys with compositions that provide resistance to hot corrosion. Alloys with high chromium content are found to resist degradation by hot corrosion very strongly (see Figure 3.48), and thus chromia- rather than alumina-forming alloys are to be preferred when corrosive conditions are prevalent. Although the degradation mechanisms of hot corrosion and sulphidisation should be distinguished, this finding is consistent with studies carried out by Mrowec *et al.* [43] on binary Ni–Cr alloys exposed to sulphur vapour at 1 atm in the range 600 to 900 °C. Below ~2 wt%Cr, the parabolic sulphidisation rate was found to be marginally greater than that of pure nickel, the scale being identified as nickel sulphide with ~0.4 wt%Cr dissolved in it; however, for greater chromium concentrations, the rate of sulphidisation was substantially less than that of pure Ni, and indeed, for Cr concentrations above 20 wt%Cr, the resistance was found to be greater than for pure Cr provided that the temperature is below 800 °C; see Figure 3.49. When the chromium content reached 50 wt%, chromium sulphide replaced nickel sulphide as the majority phase in the scale, the latter now being dispersed

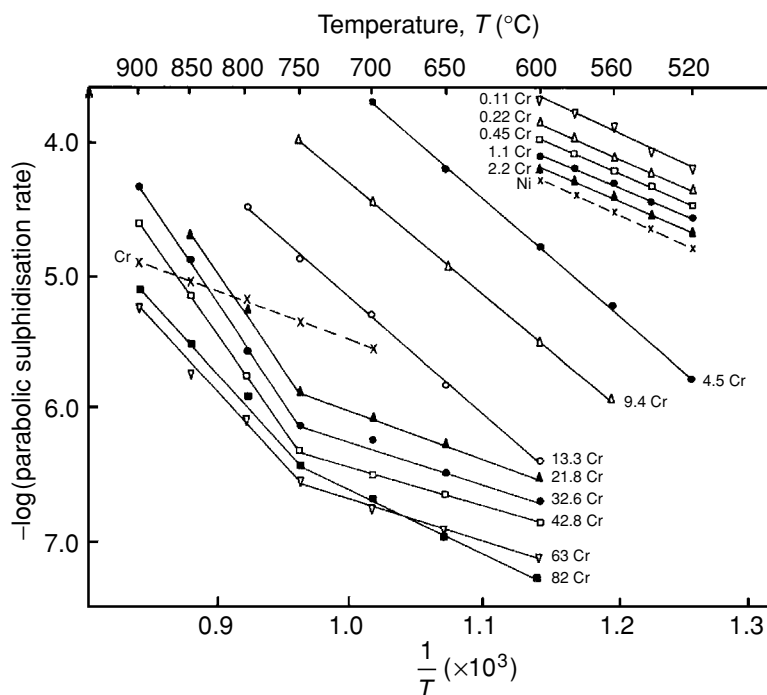


Fig. 3.49. Dependence of the parabolic sulphidisation rate for binary Ni-Cr alloys as a function of the reciprocal of the absolute reaction temperature and the chromium content [43].

in the former. Thus, resistance to both sulphidisation as well as hot corrosion is imparted by progressive additions of chromium to the superalloys. Unfortunately, the preference for high Cr content is at odds with the very best oxidation resistance, which demands an alumina former and hence a high aluminium content. A further significant factor is that Cr, being a significant γ former, is incompatible with alloys which have the very greatest creep resistance – since this requires a high γ' fraction. Furthermore, excess Cr content impairs alloy stability by promoting TCP formation. These considerations emphasise that trade-offs must be made when designing alloys for turbine applications, to achieve an optimal combination of resistances to oxidation, corrosion and creep.

3.3 Mechanical behaviour of the single-crystal superalloys

Our considerations so far have emphasised that significant stresses are set up in the turbine blading during operation. The mechanical performance of the single-crystal superalloys is then quite critical to the successful operation of the turbine engine. In this section, the response to the two most important modes of loading are considered: creep and fatigue. Since it is of the greatest practical significance, attention is restricted to the performance under uniaxial loading with the stress axis aligned along the $\langle 001 \rangle$ crystallographic direction.

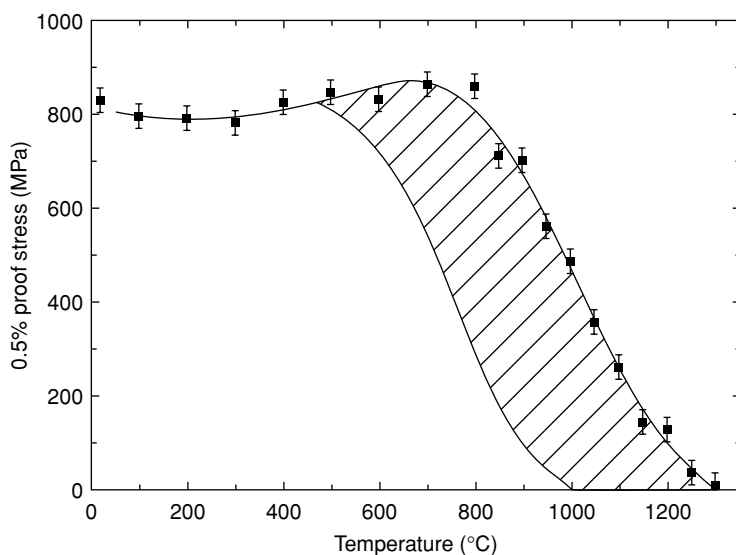


Fig. 3.50. Variation of the uniaxial 0.5% yield stress of (001)-orientated single-crystal superalloy CMSX-4, showing (hatched region) the conditions under which significant inelastic creep deformation is observed.

3.3.1 Performance in creep

In Chapter 2, it was seen that the yield stress of a typical (001)-oriented single-crystal superalloy remains rather constant from ambient conditions to approximately 700 °C. It falls to substantially lower values if the temperature is raised any further; however, in practice, of greater significance is the possibility of *time-dependent* inelastic deformation at a stress substantially lower than that required for yielding. This process is known as *creep deformation*. It is instructive to consider some numbers, for example for the second-generation single-crystal superalloy CMSX-4. The yield stress at 750 °C is about 900 MPa, yet a stress of 750 MPa is sufficient to cause a creep strain of 5% in about 5 h. The yield stress at 950 °C is about 600 MPa, but a stress of 200 MPa is sufficient for a creep strain of 5% after 1800 h. Beyond 1100 °C the yield stress falls to very low values, and in practice it is difficult to measure with accuracy because creep deformation occurs rapidly such that a substantial strain-rate dependence is displayed; however, at a temperature of 1150 °C, a stress of only 100 MPa causes 1% strain within 200 h. In Figure 3.50, yield stress data for CMSX-4 are plotted along with the values of stress at which substantial creep deformation is observed. It is clear that temperatures and stresses must be restricted if excessive creep deformation is not to occur. This is of practical significance, since components such as turbine blades are machined such that blade tip/seal clearances satisfy tight tolerances, to prevent engine performance being compromised. Typically, a set of blades must be removed from service if creep strains of greater than a few per cent are accumulated.

The mechanism of creep deformation in (001)-oriented single-crystal superalloys is very sensitive to the conditions of temperature and stress that are acting. A number of regimes can be identified, and in each a distinct mode of microstructural degradation is predominant.

Data [44] for isothermal creep performance confirms this (see those for CMSX-4, which are given in Figure 3.51); other single-crystal superalloys display similar trends, although the absolute values will be different. The data are plotted in two ways: creep strain vs. time and creep strain rate vs. creep strain – the latter is helpful since it allows periods of creep hardening to be identified, during which the creep strain rate *decreases* with *increasing* creep strain. Consider first the data at the intermediate conditions of 950 °C and 185 MPa. This type of behaviour (termed ‘tertiary’ creep deformation) is observed in all known single-crystal superalloys across a range of intermediate temperatures and stress levels, and is characterised by a creep strain rate which increases monotonically with creep strain. Completely absent is any period of ‘steady-state’ creep in which the creep strain rate is constant – as commonly observed in polycrystalline engineering alloys. At lower temperatures, and provided that the stress levels are sufficiently high (see, for example, the data at 750 °C and 750 MPa), a substantial period of ‘primary’ creep deformation occurs, with the creep strain rate first increasing and then decreasing with increasing creep strain. Thereafter, the primary creep regime gives way to a ‘secondary’ creep regime, in which the creep strain rate is invariant to a first approximation; for the data in Figure 3.51(a), the majority of the creep life is spent in the secondary regime, during which time the secondary creep rate is of order 10^{-7} /s. It is emphasised that such ‘secondary’ creep occurs only after a substantial period of primary strain, and that there have been no observed cases of single-crystal alloys which proceed straight to the ‘secondary’ regime. Finally, at temperatures beyond about 1050 °C, the creep curve displays a distinct plateau before the creep strain increases catastrophically, with rupture occurring almost immediately afterwards. There is a creep-hardening effect at first, before the plateau, during which the creep strain does not vary strongly with time. At these high temperatures, the creep behaviour is affected by a degradation of the γ/γ' microstructure, which gives rise to a ‘rafted’ morphology; one refers to a ‘rafting regime’ to distinguish this distinct behaviour from the ‘tertiary’ and ‘primary’ regimes arising at lower temperatures and greater stresses. The conditions of stress and temperature required for the operation of the primary, tertiary and rafting regimes in CMSX-4 are summarised in Figure 3.52, which is taken from ref. [45].

The micromechanisms of deformation in the different regimes are now discussed in detail.

A Tertiary creep regime

In the intermediate, ‘tertiary’, creep regime, evidence from transmission electron microscopy [26,46–49] indicates that dislocation activity is of the $a/2(1\bar{1}0)\{111\}$ type, and is restricted to the γ channels between the γ' particles (see Figure 3.53). The density of dislocations is found to increase with increasing creep strain, while other forms of microstructural degradation (γ' coarsening, creep cavitation) are absent. The γ' particles remain intact, i.e. they are not sheared by the creep dislocations; therefore, substantial cross-slip is required for substantial deformation to occur, since the γ' fraction is high and consequently the γ channels too thin to allow the dislocations to percolate the γ/γ' structure if this were not the case. The leading segments of the dislocation loops are screw in character, consistent with the requirement for cross-slip; as they expand through γ matrix channels, segments

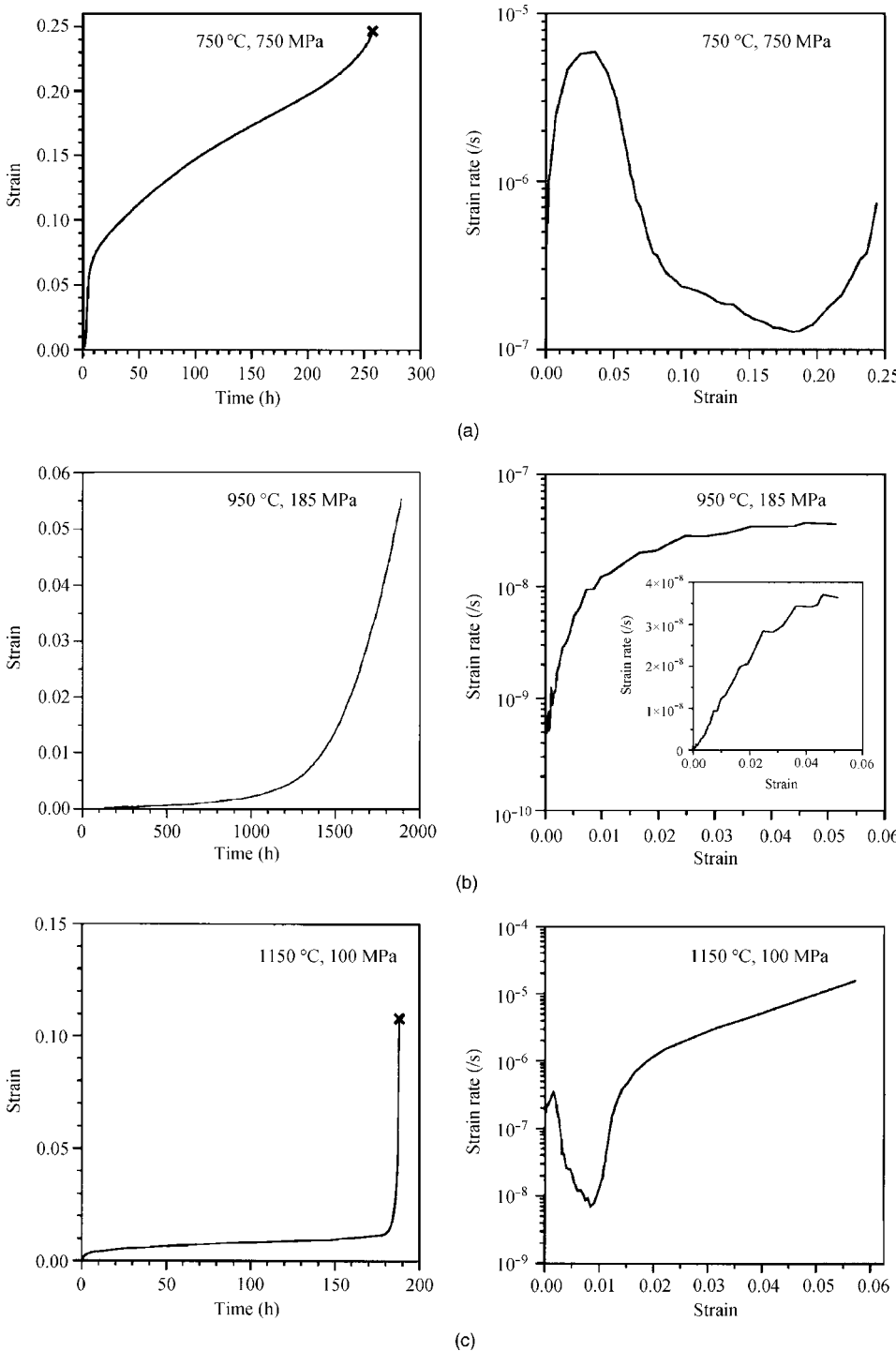


Fig. 3.51. Constant load creep data [45,46] for $\langle 001 \rangle$ -oriented single-crystal superalloy CMSX-4 at various temperatures and stresses: (a) 750 °C and 750 MPa, (b) 950 °C and 185 MPa, (c) 1150 °C and 100 MPa.

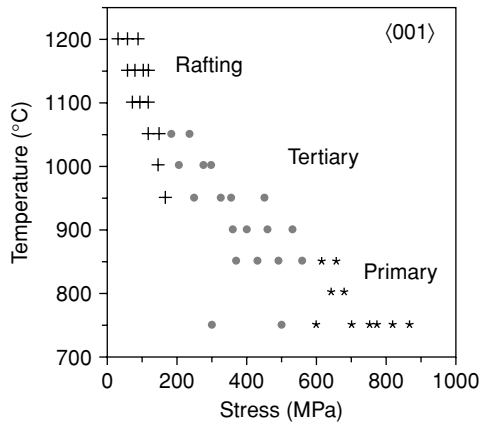


Fig. 3.52. Plot of temperature vs. applied stress [45], showing the conditions under which the primary, tertiary and rafting modes of creep deformation are operative for CMSX-4.

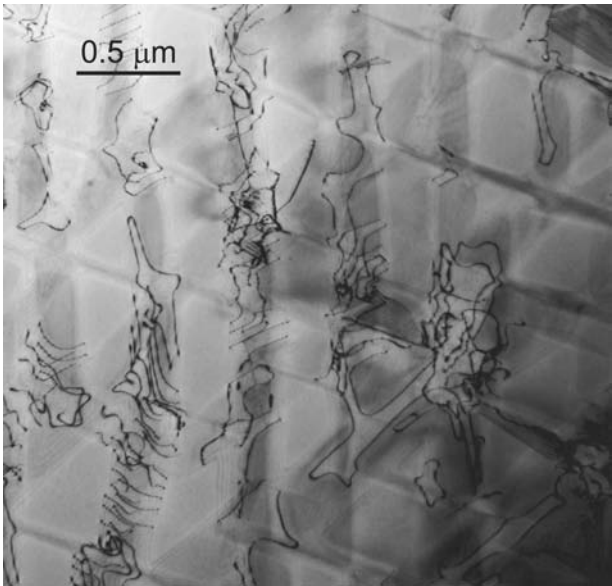


Fig. 3.53. Transmission electron micrograph of the CMSX-4 single-crystal superalloy, deformed to 0.04% strain in 1890 h at 750 °C and 450 MPa. Note the localisation of $\langle 1\bar{1}0 \rangle \{111\}$ dislocation activity in a limited number of γ channels. The foil normal is $\{111\}$. (Courtesy of Cathie Rae.)

of mixed or edge character are deposited at the γ/γ' interfaces [49]. The macroscopic deformation is found to be relatively homogeneous, with usually two or more slip systems operative for crystal orientations near $\langle 001 \rangle$ [46,50].

For the quantitative interpretation of the creep data in the tertiary regime (see Figure 3.51(b)), a theory is available to describe the observed strain softening behaviour [51]. This has been developed on the basis of the observation that, *provided that deformation is restricted to the tertiary creep regime*, the creep strain rate, $\dot{\epsilon}$, is, to a good approximation, proportional to the accumulated macroscopic creep strain, ϵ ; see Figure 3.51(b). It is reasonable to assume that the mobile dislocation density, ρ_m , which is taken to be the variable

characterising the state of damage, varies with ϵ according to

$$\rho_m = \rho_0 + M\epsilon \quad (3.28)$$

where ρ_0 is the initial dislocation density and M is a constant describing the dislocation multiplication rate. The macroscopic creep strain rate, $\dot{\epsilon}$, is then given by combining Equation (3.28) with Orowan's equation

$$\dot{\epsilon} = \rho_m b v \quad (3.29)$$

where b is the Burgers vector and v is the dislocation velocity. Provided v is constant, the proportionality of $\dot{\epsilon}$ and ϵ follows from the substitution of Equation (3.28) into Equation (3.29), yielding

$$\dot{\epsilon} = \dot{\Gamma}_i + \Omega\epsilon \quad (3.30)$$

where $\dot{\Gamma}_i$ is the initial strain rate. Calculations indicate that each of Ω and $\dot{\Gamma}_i$ shows an Arrhenius-type dependence upon temperature, consistent with

$$\Omega = a \exp \left\{ b\sigma - \frac{Q_1}{RT} \right\} \quad (3.31)$$

and

$$\dot{\Gamma}_i = c \exp \left\{ d\sigma - \frac{Q_2}{RT} \right\} \quad (3.32)$$

where a , b , c and d and Q_1 and Q_2 are constants and R is the gas constant. In the tertiary creep regime, Equations (3.28)–(3.32) have been found to model the macroscopic creep curves of CMSX-4 with reasonable accuracy (see Figure 3.54), although it is seen that the creep performance at the highest temperatures is underpredicted; more will be said about this later in the context of the rafting effect. It follows from the model that, at constant temperature and stress, the creep strain is proportional to $\exp\{kt\}$, where k is a constant and t is time; therefore it increases exponentially with time t .

The model ignores one important aspect of the deformation behaviour in the tertiary regime: the incubation period during which $a/2(1\bar{1}0)\{111\}$ dislocations multiply from dislocations sources, typically low-angle grain boundaries, into regions which are initially dislocation-free [26]. Particularly when the temperature is low, the creep strain rate is very small and the creep strain is less great than predicted by an interpolation from the behaviour at longer times, using Equations (3.28)–(3.32). During the incubation period, the TEM evidence indicates that the glide of dislocations occurs preferentially through the horizontal γ channels, i.e. the channels which lie perpendicular to the direction of the applied stress. This can be rationalised in the following way: the superposition of the (negative) misfit stress for a negatively misfitting superalloy and the externally applied (positive) tensile stress along $\langle 001 \rangle$ is such as to restrict dislocation activity in the vertical channels; conversely, the stress in the horizontal channels is, to a first approximation, unaffected. Detailed calculations which take account of the Poisson effect and a determination of the misfit stress indicate that the horizontal channels experience a shear stress at least twice as great as that in the vertical channels [26,52]. However, the incubation period is short under most circumstances,

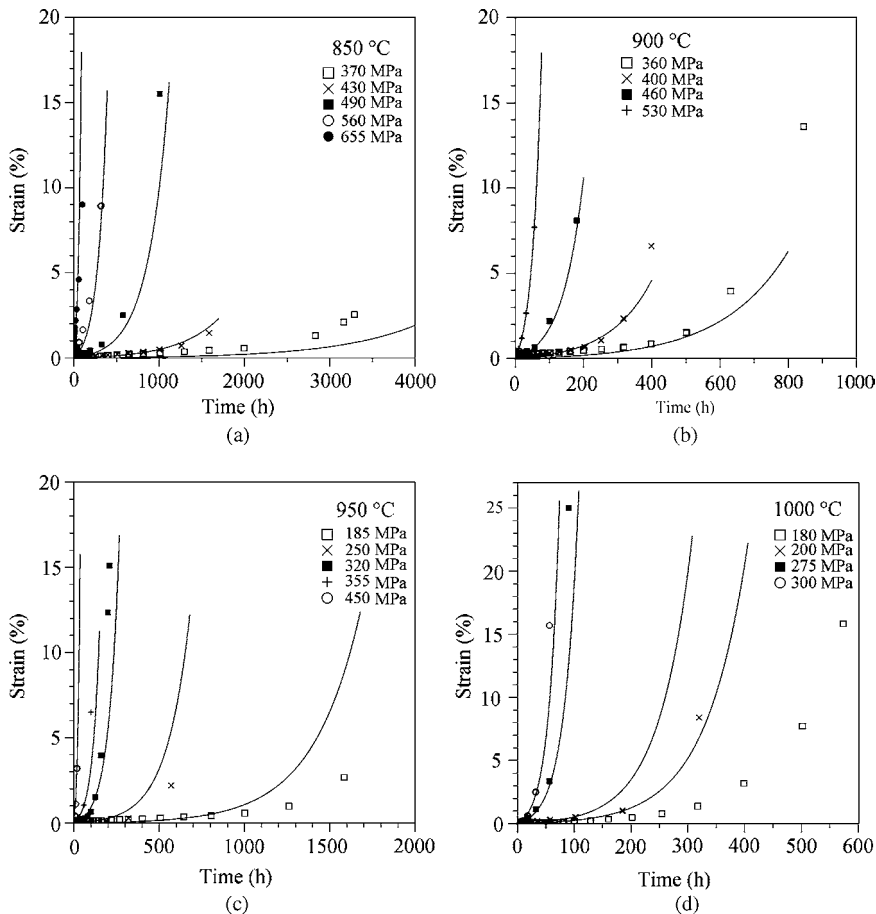


Fig. 3.54. A comparison of the creep data for the CMSX-4 single-crystal superalloy and the tertiary creep model described by Equations (3.28)–(3.32). Note that the agreement between model and experiment is generally good, except at high temperatures and low stresses. Data from refs. [45] and [46].

and the equations given above can be used to describe the creep strain evolution in the tertiary regime with reasonable accuracy.

B Primary creep regime

At low temperatures, and provided the applied stress is sufficiently high, $\langle 001 \rangle$ -oriented single-crystal superalloys undergo significant primary creep deformation. In Figure 3.55, creep curves for CMSX-4 are plotted at temperatures of 750 °C and 850 °C for specimens of identical orientation and within a few degrees of $\langle 001 \rangle$. A number of observations can be made. First, the primary creep strain increases with applied stress, with primary creep eventually giving way to secondary creep soon after the knee of each curve; thus, significant primary creep strain occurs only when the applied stress is greater than a critical threshold

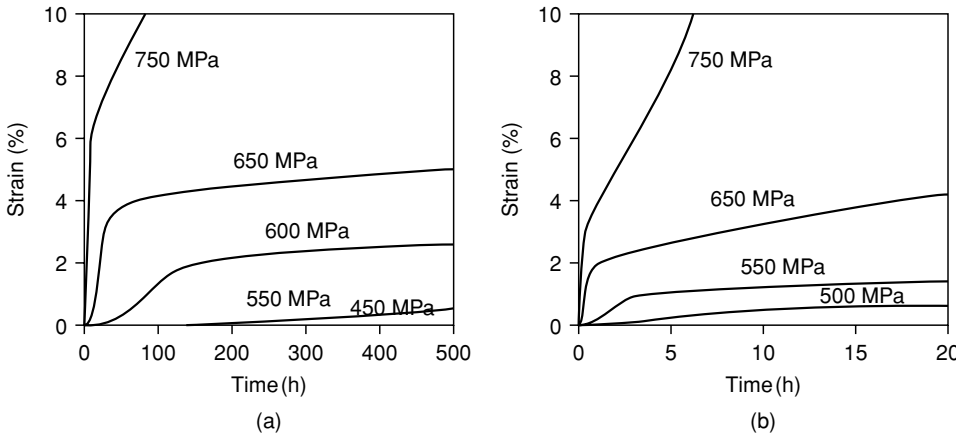


Fig. 3.55. Constant load creep data for CMSX-4 at low temperatures and high stresses, i.e. in the primary creep regime. (a) 750 °C; (b) 850 °C.

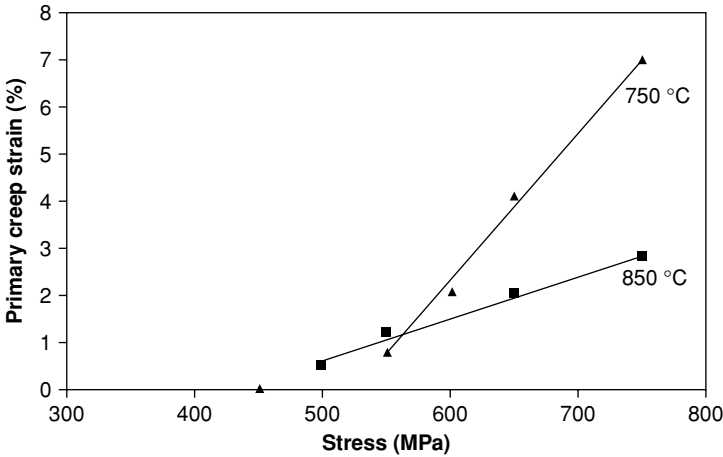


Fig. 3.56. Plot of percentage primary creep strain against applied stress, confirming a threshold stress of ~ 500 MPa for operation of the primary creep mechanism.

value of about 500 ± 50 MPa; see Figure 3.56. Second, the transition from primary to secondary creep occurs at a time which decreases as the applied stress increases; moreover, further analysis reveals that the secondary creep rate scales in proportion to the primary creep strain [46].

A characteristic of primary creep is a significant heterogeneity of slip. If the creep specimen is cylindrical in shape, as is most usually the case, then after creep deformation the cross-section is found to be elliptical (see Figure 3.57), with the degree of ellipticity increasing with the primary creep strain incurred. More significantly, careful characterisation of the shape change and correlation of it with the specimen orientation indicates that the macroscopic shape change is consistent with $\langle 11\bar{2} \rangle \{111\}$ deformation [46,53]. This observation was first made by Kear and co-workers in the late 1960s; they proposed a model for

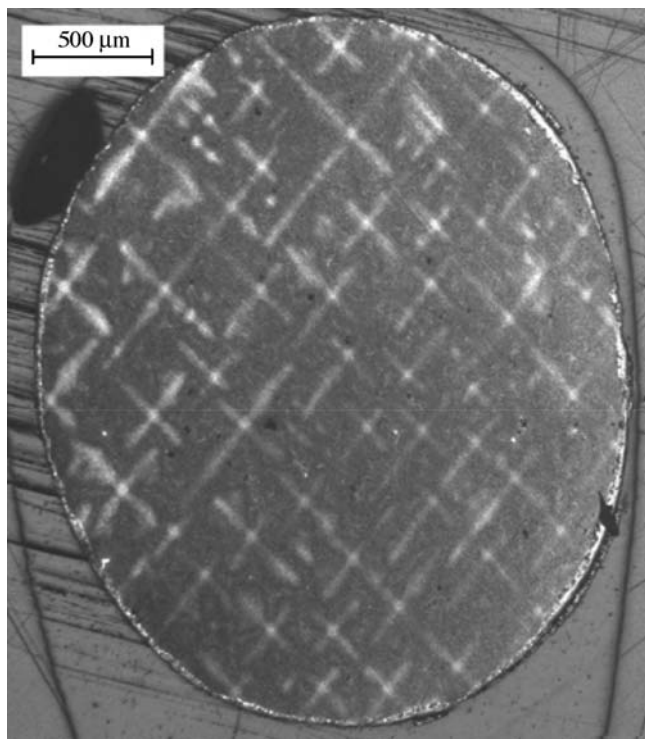
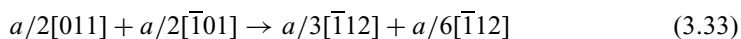


Fig. 3.57. Cross-section of a single-crystal creep specimen deformed in the primary creep regime [53], illustrating the ellipticity caused by severe non-uniform deformation.

$\langle 11\bar{2} \rangle \{111\}$ slip, which is supported by TEM studies, indicating that deformation occurs by the movement of dislocation ribbons of net Burgers vector $a\langle 11\bar{2} \rangle$ [54]. These are able to shear the γ' precipitates in a conservative fashion. Since the Burgers vector of the ribbon is large at $a\sqrt{6}$, significant amounts of strain are accumulated as the ribbons sweep through the γ/γ' structure. If the applied stress is below the threshold for primary creep to operate, then the microscopic evidence shows that only dislocations of form $a/2\langle 1\bar{1}0 \rangle \{111\}$ are present, and that these reside in γ . This is very strong evidence that the $a\langle 11\bar{2} \rangle$ ribbons are responsible for primary creep. Figure 3.58 shows a transmission electron micrograph of an $a\langle 11\bar{2} \rangle$ ribbon in a creep-deformed TMS-82 single-crystal superalloy.

There has been considerable debate about the mechanism of formation of the $a\langle 11\bar{2} \rangle$ dislocation ribbons, but it is now known to involve the reaction of $a/2\langle 1\bar{1}0 \rangle \{111\}$ dislocations in γ , which are dissociated into their Shockley partials. A typical reaction might then be given by [55]



If the applied stress is sufficient, the $a/3[\bar{1}12]$ dislocation is able to enter the γ' , leaving a superlattice intrinsic stacking fault (SISF) behind it (see Figure 3.59(a)) and the remaining $a/6[\bar{1}12]$ at the γ/γ' interface. Formation of the ribbon in γ' then requires further reactions between the $a/2\langle 1\bar{1}0 \rangle \{111\}$ -type dislocations, such that the original $a/6[\bar{1}12]$, a further $a/6[\bar{1}12]$ and eventually a second $a/3[\bar{1}12]$, enter the γ' ; see Figure 3.59(b). An anti-phase

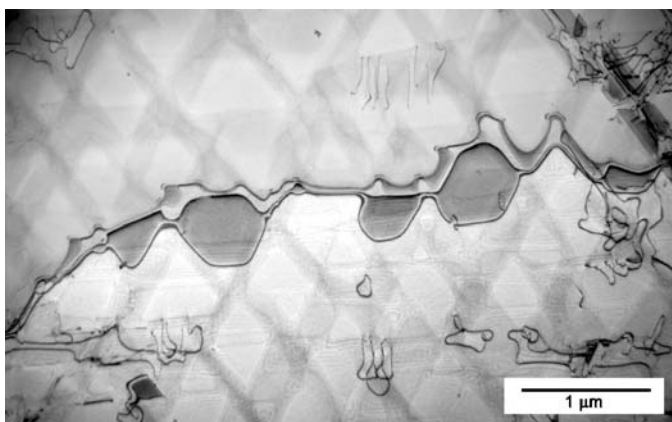


Fig. 3.58. TEM micrograph of an $a\langle 11\bar{2} \rangle$ ribbon in the single-crystal superalloy TMS-82, deformed in creep at 750 °C and 750 MPa to 11% strain. The foil normal is {111}. (Courtesy of Cathie Rae.)

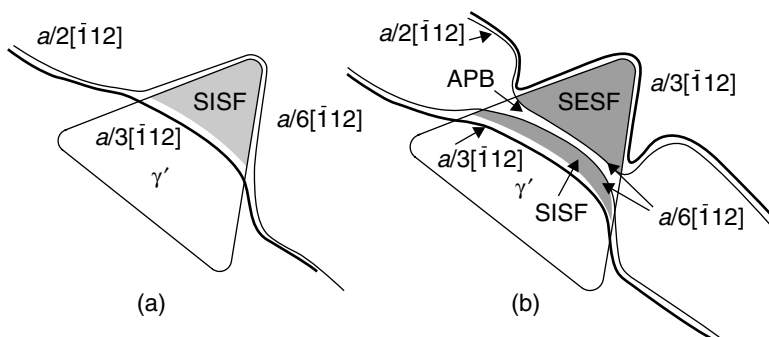


Fig. 3.59. Illustration of the mechanism of shearing of the γ' by an $a\langle 11\bar{2} \rangle$ ribbon during primary creep. An $a/3[\bar{1}12]$ dislocation enters the γ' , leaving a superlattice intrinsic stacking fault (SISF) behind it, (a), leaving an $a/6[\bar{1}12]$ partial at the γ/γ' interface. In (b), the original $a/6[\bar{1}12]$ and a further $a/6[\bar{1}12]$ have entered the γ' , leaving an $a/3[\bar{1}12]$ at the interface.

boundary (APB) then separates the two $a/6[\bar{1}12]$ dislocations, and a superlattice extrinsic stacking fault (SESF) the second $a/6[\bar{1}12]$ and the final $a/3[\bar{1}12]$. Once formed, the ribbon is able to glide through the γ' precipitates in a glissile fashion. Ribbons intersecting a γ' precipitate cause it to be populated with stacking faults. The cessation of primary creep occurs when the $a\langle 11\bar{2} \rangle$ dislocation ribbon is unable to glide conservatively through the γ/γ' structure, primarily due to the adsorption of $a/2\langle 110 \rangle\{111\}$ dislocation segments at the γ/γ' interfaces.

A consequence of the mechanism by which primary creep occurs in single-crystal superalloys is a considerable creep anisotropy. By this one means that the creep life is affected very strongly by small changes in the orientation of the tensile axis with respect to the $\langle 001 \rangle$ crystallographic axis. This was noted by MacKay and Maier [56], who plotted the variation of the creep life of Mar-M247 single-crystals deformed at 760 °C on a $\langle 001 \rangle/\langle 011 \rangle/\langle 111 \rangle$ standard triangle of the stereographic projection; see Figure 3.60. Misorientations of only 10° from $\langle 001 \rangle$ were found to have a profound effect on creep life, with orientations near

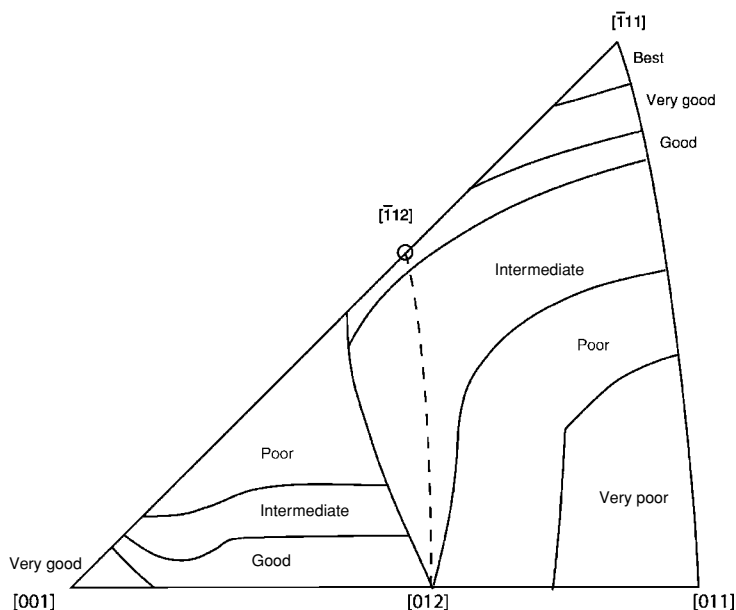


Fig. 3.60. Variation of the performance of the Mar-M247 single-crystal superalloy deformed in primary creep at 760 °C. Orientations near $\langle 011 \rangle$ have the shortest rupture lives, orientations near $\langle 111 \rangle$ the best, with $\langle 001 \rangle$ somewhere between. After ref. [56].

to the $\langle 001 \rangle / \langle 011 \rangle$ boundary displaying very much better performance than those near to the $\langle 001 \rangle / \langle 111 \rangle$ one. While the details are beyond the scope of this book, these effects arise because the movement of the $a\langle 11\bar{2} \rangle$ ribbon requires (i) sufficient densities of at least two $a/2\langle 1\bar{1}0 \rangle \{111\}$ families of dislocations in γ , to enable the nucleation of the dislocation ribbons in accordance with the reaction given above, and (ii) sufficient resolved shear stress for the propagation of the $a\langle 11\bar{2} \rangle$ ribbon. Thus conditions for both the nucleation and the propagation of the ribbon must be satisfied if appreciable primary creep is to occur [57]. Given this remarkably complex mechanism of primary creep deformation in single-crystal superalloys, it is unsurprising that its extent is remarkably sensitive to the chemical composition, with cobalt and rhenium in particular promoting primary creep and tantalum suppressing it [55,57,58]; it seems plausible that stacking fault shear of γ' is promoted by HCP elements such as Co and Re since they lower the stacking fault energy in γ . This suggestion is consistent with observations that primary creep in Co-containing single-crystal superalloys is highly sensitive to orientation, with creep anisotropy being suppressed as the Co content is reduced [58].

C Rafting regime

In the rafting regime, the γ/γ' microstructure degrades rather quickly because thermally activated processes are favoured strongly since the temperature is high. Two effects occur at a very early stage of deformation: (i) equilibrium interfacial dislocation networks form at the γ/γ' interfaces [59], and (ii) the γ' particles coalesce by a process of directional coarsening

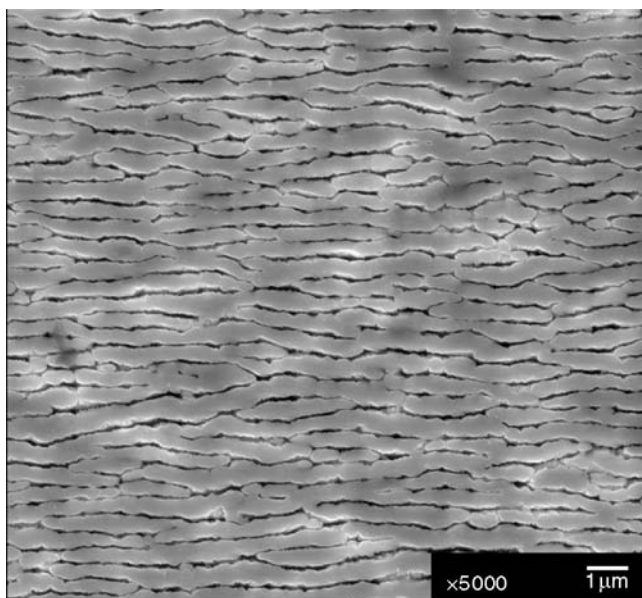


Fig. 3.61. Rafted γ/γ' structure in the CMSX-4 superalloy, deformed at 1150 °C and 100 MPa to 0.39% strain in 10 h [62].

known as the ‘rafting’ effect [49,60–62]. Figure 3.61 illustrates the rafted morphology in the CMSX-4 superalloy after tensile creep loading to 0.39% strain in 10 h at 1150 °C and 100 MPa [62]. The axis of loading lies in the plane of the micrograph in the vertical direction; thus the normals to the broad faces of the rafts – which are plate-shaped in form – lie perpendicular to the axis of loading. This is always found to be the case for negatively misfitting alloys such as CMSX-4, into which category most commercially developed alloys fall. When the misfit is positive, needle-shaped rafts are found whose long axes lie parallel to the direction of loading. Changing the sign of the stress, i.e. compressive rather than tensile loading, has an effect on the raft morphology which is equivalent to changing the sign of the misfit.

The interfacial dislocation networks arise from $a/2\langle 1\bar{1}0 \rangle\{111\}$ creep dislocations originating in the γ phase; these become captured by the γ/γ' interfaces, and the networks are formed by a series of dislocation reactions. The dislocations are primarily of edge character with the extra half-planes in the γ' phase if $\delta < 0$. Thus by this mechanism the perfect elastic coherency of the interfaces is lost and misfit stresses are relieved. Figure 3.62 is taken from the work of Field, Pollock and co-workers [63], who have analysed the networks in detail using TEM microscopy – this specimen was interrupted after 20 h with 0.5% creep strain having been accumulated at 1093 °C. Note in particular the square configurations of four dislocations of Burgers vector of $a/2\langle 1\bar{1}0 \rangle$ type; they are of edge character and are thus efficient at relieving the misfit, but these segments have no shear stress resolved on them – thus they must have been formed by reactions driven by reductions in misfit and dislocation line energies. A model has been proposed by these authors which accounts for their formation; see Figure 3.63. Initially, on a γ/γ' interface which has $[001]$ as its

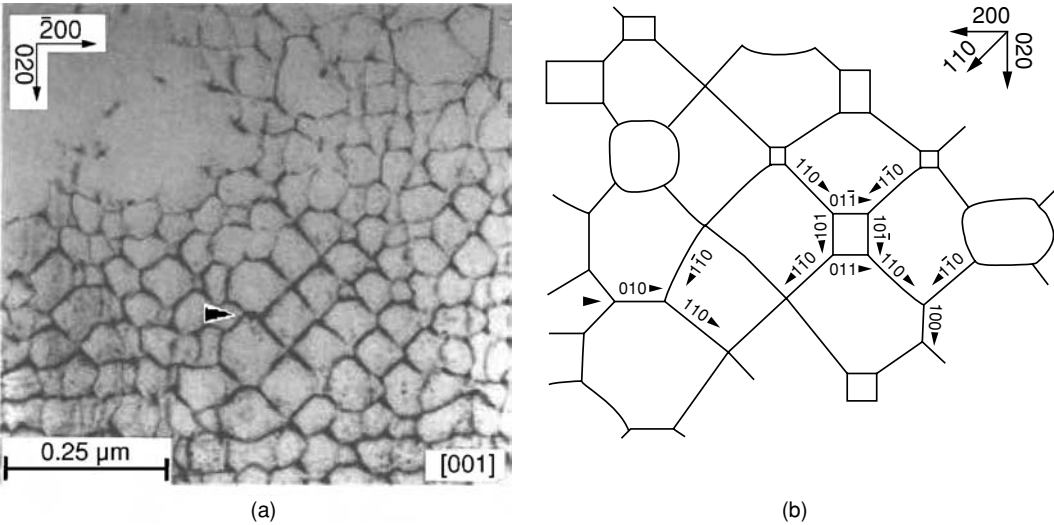


Fig. 3.62. (a) TEM micrograph and (b) accompanying schematic interpretation of an experimental single-crystal superalloy deformed to 0.5% creep strain in 20 h at 1093°C , showing dislocation networks at the γ/γ' interfaces; arrows in (a) and (b) show equivalent positions. Foil normal $\{001\}$, equivalent to the direction of loading. After ref. [63].

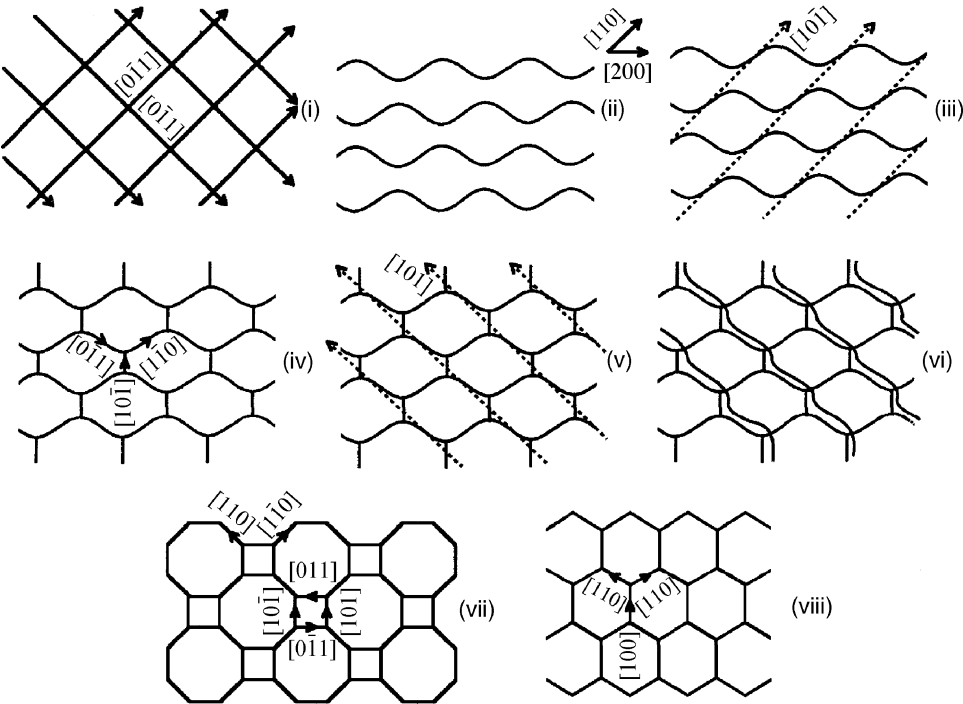


Fig. 3.63. Schematic representation of dislocation reactions leading to the formation of mismatch accommodating nets on the (001) γ/γ' interfaces during creep. After ref. [63].